A Failure of Π^1_{n+3}-Reduction in the Presence of Σ^1_{n+3}-Separation

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Abstract

We show that one can force over L that Σ^1_3-separation holds, while Π^1_3-reduction fails, thus separating these two principles for the first time. The construction can be lifted to canonical inner models M_n with n-many Woodin cardinals, yielding that assuming the existence of M_n, Σ^1_{n+3}-separation can hold, yet Π^1_{n+3}-reduction fails.

1 Introduction

Descriptive Set Theory serves as a fundamental framework for investigating the structure and properties of sets of real numbers. Two central concepts within this theory, the Separation Property, introduced in the early 1920's and the Reduction Property, introduced by Kuratowski in the mid 1930's, have garnered significant attention due to their profound implications for properties of projective subsets of the real numbers.

Definition 1.1. *We say that a projective pointclass* $\Gamma \in \{\mathbf{\Sigma^1_n} \mid n \in \omega\} \cup \{\mathbf{\Pi^1_n} \mid n \in \omega\} \cup \{\Sigma^1_n \mid n \in \omega\} \cup \{\Pi^1_n \mid n \in \omega\}$ *has the separation property (or just separation) iff every pair* A_0 *and* A_1 *of disjoint elements of* Γ *has a separating set* C*, i.e. a set* C *such that* $A_0 \subset C$ *and* $A_1 \subset \omega^\omega \backslash C$ *and such that* $C \in \Gamma \cap \check{\Gamma}$*, where* $\check{\Gamma}$ *denotes the dual pointclass of* Γ*.*

Definition 1.2. *We say that a projective pointclass* $\Gamma \in \{\mathbf{\Sigma^1_n} \mid n \in \omega\} \cup \{\mathbf{\Pi^1_n} \mid n \in \omega\} \cup \{\Sigma^1_n \mid n \in \omega\} \cup \{\Pi^1_n \mid n \in \omega\}$ *satisfies the* Γ*-reduction property (or just reduction) if every pair* B_0, B_1 *of* Γ*-subsets of the reals can be reduced by a pair of* Γ*-sets* R_0, R_1*, which means that* $R_0 \subset B_0$*,* $R_1 \subset B_1$*,* $R_0 \cap R_1 = \emptyset$ *and* $R_0 \cup R_1 = B_0 \cup B_1$*.*

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It follows immediately from the definitions that Γ-reduction implies $\check{\Gamma}$-separation. It is very natural to ask whether the reverse direction is also true.

Since their introduction, many results have been proved which shed light on how separation and reduction can behave among the projective point-classes. These results can be obtained using two very different set theoretic assumptions, which draw very different scenarios of the properties of separation and reduction.

The first assumption is $V = L$, or rather the existence of a Σ^1_2-definable, good projective well-order of the reals. Recall that a good Σ^1_2-definable well-order is a Σ^1_2-well-order with the additional property that also the relation $\mathrm{InSeg}(x, y) :\Leftrightarrow \{(x)_i \mid i \in \omega\} = \{z \mid z <_L y\}$ is Σ^1_2, where $(x)_i$ denotes some recursive decoding of x into ω-many reals, $(x)_i$ being the i-th real decoded out of r. By results of J. Addison ([1]) the existence of a good Σ^1_n-wellorder implies Σ^1_m-uniformization for every $m \geqslant n$. As Σ^1_m-uniformization implies Σ^1_m reduction, the assumption of $V = L$ implies that for every $n \geqslant 1$, Σ^1_n-reduction and Π^1_n-separation is true (the case $n = 1$ follows from Kondo's theorem that Π^1_1-uniformization is true).

The second assumption which settles the behaviour of reduction and separation on the projective hierarchy is projective determinacy (PD). By the results of Y. Moschovakis ([14]), under PD, for every $n \in \omega$, Π^1_{2n+1} and Σ^1_{2n+2} sets have the scale property, which in particular implies that Π^1_{2n+1} and Σ^1_{2n+2} sets have the uniformization property and so Π^1_{2n+1} and Σ^1_{2n+2} reduction is true. By the famous theorems of D. Martin and J. Steel (see [12]) on the one hand, and H. Woodin (see [11]) on the other hand, determinacy assumptions on projective sets and large cardinal assumptions are two sides of the very same coin.

Note that under $V = L$ and also under PD, Γ separation holds because already the stronger $\check{\Gamma}$-reduction (in fact $\check{\Gamma}$-uniformization) holds. So these results do not shed light on the question stated above, whether Γ-separation and $\check{\Gamma}$-reduction are different properties at all. A partial answer to the question was first given by R. Sami in his PhD thesis from 1976 [15]. In it he showed (among other interesting results) that after adding a single Cohen real to L, the resulting universe will satisfy that Π^1_3-separation holds, yet Σ^1_3-reduction fails. And additionally Σ^1_n-reduction holds again for $n \geqslant 4$. His results inspired L. Harrington to produce a model in which $\mathbf{\Pi^1_3}$-separation holds but there is a (lightface) Σ^1_3 set, which can not be reduced by any pair of $\mathbf{\Sigma^1_3}$-sets, thus Σ^1_3-reduction fails (a write up of Harrington's proof can be found in [10]).

The question for the other side of the projective hierarchy, namely for Σ^1_3-separation and Π^1_3-reduction remained open though since then.

Goal of this article is to produce the counterpart to L. Harrington's result:

Theorem. *One can force over L a model of $\mathbf{\Sigma^1_3}$-separation over which there*

is a pair of Π^1_3*-sets, which can not be reduced by any pair of* $\mathbf{\Pi^1_3}$*-sets.*

We add that our result uses completely different techniques and methods than Sami's and Harrington's theorems. It follows from a modification of the arguments from [7], that the proofs can be transferred from L to M_n, the canonical inner model with n-many Woodin cardinals.

Theorem. *Assuming that* M_n *exists, there is a model of* $\mathbf{\Sigma^1_{n+3}}$*-separation over which there is a pair of* Π^1_{n+3}*-sets, which can not be reduced by any pair of* $\mathbf{\Pi^1_{n+3}}$*-sets.*

The proof relies on the construction and ideas from [5], where a universe with the $\mathbf{\Sigma^1_3}$-separation property is produced via forcing over L. However, we will introduce some simplifications of the original argument, yielding a cleaner presentation. We will use the coding machinery form [6] which is basically the same as in [7]. In [6] a universe is forced over L, where Π^1_3-reduction holds but the stronger Π^1_3-uniformization fails. Our proof presented here will take a quite different direction though and uses a more direct diagonalization argument, where we actively work towards two Π^1_3-sets B_0, B_1 which can not be reduced by a pair of $\mathbf{\Pi^1_3}$-sets. As we simultaneously have to work towards a stronger failure of Π^1_3-reduction, we need to substantially alter the original definitions and arguments for forcing the $\mathbf{\Sigma^1_3}$-separation.

1.1 Notation

The notation we use will be mostly standard, we hope. Diverging from the conventions we write $\mathbb{P} = (\mathbb{P}_\alpha : \alpha < \gamma)$ for a forcing iteration of length γ with initial segments $\mathbb{P}_\alpha$. The α-th factor of the iteration will be denoted with $\mathbb{P}(\alpha)$, this is nonstandard as typically one writes $\dot{\mathbb{Q}}_\alpha$. Note here that we drop the dot on $\mathbb{P}(\alpha)$, even though $\mathbb{P}(\alpha)$ is in fact a $\mathbb{P}_\alpha$-name of a partial order. If $\alpha' < \alpha < \gamma$, then we write $\mathbb{P}_{\alpha'\alpha}$ to denote the intermediate forcing of $\mathbb{P}$ which happens in the interval $[\alpha', \alpha)$, i.e. $\mathbb{P}_{\alpha'\alpha}$ is such that $\mathbb{P} \cong \mathbb{P}_{\alpha'} * \mathbb{P}_{\alpha'\alpha}$.

We write $\Sigma_n(X)$, for X an arbitrary set, to denote the set of formulas which are Σ_n and use elements from X as a parameter.

We write $\mathbb{P} \Vdash \varphi$ whenever every condition in $\mathbb{P}$ forces φ, and make deliberate use of restricting partial orders below conditions, that is, if $p \in \mathbb{P}$ is such that $p \Vdash \varphi$, we let $\mathbb{P}' := \mathbb{P}_{\leqslant p} := \{q \in \mathbb{P} : q \leqslant p\}$ and use $\mathbb{P}'$ instead of $\mathbb{P}$. This is supposed to reduce the notational load of some definitions and arguments. We also write $V[\mathbb{P}] \models \varphi$ to indicate that for every $\mathbb{P}$-generic filter G over V, $V[G] \models \varphi$, and use $V[\mathbb{P}]$ to denote the generic extension of V by $\mathbb{P}$ in case the particular choice of the generic filter does not matter in the current context.

2 Independent Suslin trees in L, almost disjoint coding

The coding method of our choice utilizes Suslin trees, which can be generically destroyed in an independent way of each other.

Definition 2.1. *Let $\vec{T} = (T_\alpha : \alpha < \kappa)$ be a sequence of Suslin trees. We say that the sequence is an independent family of Suslin trees if for every finite set of pairwise distinct indices $e = \{e_0, e_1, ..., e_n\} \subset \kappa$ the product $T_{e_0} \times T_{e_1} \times \cdots \times T_{e_n}$ is a Suslin tree again.*

Note that an independent sequence of Suslin trees $\vec{T} = (T_\alpha : \alpha < \kappa)$ has the property that if $A \subset \kappa$ and we form $\prod_{i \in A} T_i$ with finite support, where each T_i denotes the forcing we obtain if we force with the nodes of the tree as conditions using the tree order as the partial order, then in the resulting generic extension $V[G]$, for every $\alpha \notin A$, $V[G] \models$ "T_α is a Suslin tree". To see this, we assume the opposite, namely there is an $\alpha \notin A$ such that $V[G]$ thinks that T_α is not Suslin anymore. We note that $V[G]$ is a generic extension of V obtained with a forcing with the ccc (see [4], Lemma 51 for a more general argument of this type). So there is an $\aleph_1$-sized antichain of T_α in $V[G]$. But then the finitely supported product of the form $(\prod_{i \in A} T_i) \times T_\alpha$ can not have the ccc, which is a contradiction to the assumed independence of $\vec{T}$.

One can easily force the existence of independent sequences of Suslin trees with products of Jech's or Tennenbaum's forcing, or with just products of ordinary Cohen forcing. On the other hand independent sequences of length ω_1 already exist in L.

Theorem 2.2. *Assume that $\aleph_1 = \aleph_1^L$ and that $(M, \in)$ is a transitive, ω_1-containing, uncountable model of ZFC^- + "$\aleph_1$ exists". Then there is an independent sequence $\vec{T} = (T_\alpha \mid \alpha < \omega_1)$ of L-Suslin trees and the sequence $\vec{T}$ is uniformly $\Sigma_1(\{\omega_1\})$-definable over M. To be more precise, there is a Σ_1-formula ϕ with ω_1 as the unique parameter, which does not depend on the model M, such that the relation $\{(t, \gamma, \eta) \mid \gamma, \eta < \omega_1 \wedge t \in T_\eta^\gamma\}$, where T_η^γ denotes the γ-th level of $T_\eta \in \vec{T}$, is definable over M using ϕ. In particular $\vec{T}$ can be defined correctly in any transitive, uncountable model M of ZFC^- and "$\aleph_1$ exists" using ϕ and ω_1 as a parameter of ϕ.*

Proof. We first argue in L and later argue that our to be presented construction relativizes to any M as in the proposition. We fix the canonical $\diamondsuit$-sequence $(a_\alpha \subset \alpha \mid \alpha < \omega_1)$ in L. Next we alter the usual construction of a Suslin tree from $\diamondsuit$ to construct an ω_1-sequence of Suslin trees $\vec{T} = (T_\alpha \mid \alpha < \omega_1)$.

We fix a Δ_1-definable bijection of the set of pairs of countable ordinals with ω_1. We use this bijection in the usual way to code sets of pairs of

countable ordinals with sets of countable ordinals. Under this identification, trees will be coded as subsets of ordinals.

We assume that $\alpha < \omega_1$ is a limit ordinal, and split the set $a_\alpha \subset \alpha$ into two parts via considering the even and odd ordinals of a_α. We assume first that if we recursively split again the even ordinals of a_α into three sets $A, B, C \subset \mathrm{Even}(a_\alpha)$, then A codes a wellorder of ordertype $\beta < \alpha < \omega_1$, B codes a wellorder of ordertype $\gamma < \alpha$ and C codes a finite subset of ω_1. This indicates that we aim to define the β-th level of T_γ as follows.

We assume first that β is a limit ordinal and T^i_γ and T^i_η for $\eta \in C$ has been defined already for each $i < \beta$. Now we consider the odd ordinals of a_α and assume that $\mathrm{Odd}(a_\alpha)$ codes a set $A \subset \bigcup_{i<\beta} T^i_\gamma \times \prod_{\eta \in C} T^i_\eta$ which is an antichain there.

We define the β-th level T^β_γ of T_γ such that it seals off the antichain A. To be more specific we chose $T^\beta_\gamma \times \prod_{\eta \in C} T^\beta_\eta$ in such a way that A remains a maximal antichain in all further extensions of $T^\beta_\gamma \times \prod_{\eta \in C} T^\beta_\eta$.

Otherwise we just extend T^β_γ via adding top nodes on countably many branches through T^β_γ.

If β is not a limit ordinal we define T^β_γ under the assumption that $T^{\beta-1}_\gamma$ is already defined and let T^β_γ be just the result of putting ω-many new nodes above each node of $T^{\beta-1}_\gamma$.

We let $T_\gamma := \bigcup_{\beta<\omega_1} T^\beta_\gamma$ and claim that $(T_\gamma \mid \gamma < \omega_1)$ is an independent sequence of Suslin trees in L.

Indeed, if $A \in L$ is an antichain in some $\prod_{\gamma \in e} T_\gamma$, then there is a constructible club of ordinals α such that $A \cap \alpha = a_\alpha$ and $\mathrm{Even}(a_\alpha)$ codes A, B, C where A codes β, B codes γ and C codes $e \backslash \{\gamma\}$ and $\mathrm{Odd}(a_\alpha)$ codes a maximal antichain in $\bigcup_{i<\beta} T^i_\gamma \times \prod_{\eta \in C} T^i_\eta$. But then $\mathrm{Odd}(a_\alpha)$ got sealed off and $\prod_{\gamma \in e} T_\gamma$ has no uncountable antichains, so is a Suslin tree in L.

The definability of $\vec{S}$ comes from the fact that the canonical $\diamondsuit$-sequence in L is $\Sigma_1(\{\omega_1\})$-definable. We can use L_{ω_1} (which is $\Sigma_1(\{\omega_1\})$ to correctly define $\diamondsuit$ over it and consequentially $\vec{S}$ becomes definable over L_{ω_1} as well. The above considerations can be simulated correctly already in any transitive, uncountable M which models ZFC^-, as it will compute L_{ω_1} correctly and the rest of the construction is performed inside the latter model. $\square$

We remark, that the above proof shows that countable levels of countable initial segments of $\vec{S}$ are already uniformly definable over countable transitive models M of ZF^- and "$\aleph_1$ exists and $\aleph_1 = \aleph_1^L$". That is, if M is as just stated and $\gamma, \eta < \aleph_1^M$ and $t \in T^\gamma$ such that $\phi(t, \gamma, \omega_1)$ is true in L, then $M \models \phi(t \restriction \omega_1^M, \gamma, \omega_1)$ and vice versa. Indeed if M is such a model, and $M \cap \omega_1 = \aleph_1^M = (\aleph_1^L)^M$, then the canonical $\diamondsuit$-sequence can be computed correctly up to $\aleph_1^M$ inside M's version of L. So the whole construction of $\vec{S}$ inside L_{ω_1}, when repeated inside $L_{\aleph_1^M}$ will produce the claimed initial

segment of $\vec{S}$ restricted to trees of height $\aleph_1^M$.

The trees from $\vec{S}$ will later be used to define two countable families of Σ^1_3-sets, which will be manipulated in such a way that they will witness $\mathbf{\Sigma^1_3}$-separation and the failure of Π^1_3-reduction. As we need two such families, we split $\vec{S}$ into two independent sequences

$$\vec{S}^0 := \{S_\alpha \mid \alpha \text{ is even}\}$$

and

$$\vec{S}^1 := \{S_\alpha \mid \alpha \text{ is odd.}\}$$

We briefly introduce the almost disjoint coding forcing due to R. Jensen and R. Solovay. We will identify subsets of ω with their characteristic function and will use the word reals for elements of 2^ω and subsets of ω respectively. Let $D = \{d_\alpha : \alpha < \aleph_1\}$ be a family of almost disjoint subsets of ω, i.e. a family such that if $r, s \in D$ then $r \cap s$ is finite. Let $X \subset \omega$ be a set of ordinals. Then there is a ccc forcing, the almost disjoint coding $\mathbb{A}_D(X)$ which adds a new real x which codes X relative to the family D in the following way

$$\alpha \in X \text{ if and only if } x \cap d_\alpha \text{ is finite.}$$

Definition 2.3. *The almost disjoint coding $\mathbb{A}_D(X)$ relative to an almost disjoint family D consists of conditions $(r, R) \in [\omega]^{<\omega} \times D^{<\omega}$ and $(s, S) < (r, R)$ holds if and only if*

1. *$r \subset s$ and $R \subset S$.*
2. *If $\alpha \in X$ and $d_\alpha \in R$ then $r \cap d_\alpha = s \cap d_\alpha$.*

We shall briefly discuss the L-definable, $\aleph_1^L$-sized almost disjoint family of reals D we will use throughout this article. The family D is the canonical almost disjoint family one obtains when recursively adding the $<_L$-least real x_β not yet chosen and replace it with $d_\beta \subset \omega$ where this d_β is the real which codes the initial segments of x_β using some recursive bijections between ω and $\omega^{<\omega}$. The definition of D is uniform over any uncountable, transitive ZF^--models M with, as we can correctly compute L up to $\aleph_1^L$ inside M and then apply the above definition inside L's version of M. Even more is true, if M is a countable, transitive model of $\mathsf{ZF}^- +$ " $\aleph_1$ exists and $\aleph_1 = \aleph_1^L$", then M will compute $D \restriction \omega_1^M$ in a correct way. The reason is again, that M can define an initial segment of L correctly which suffices to calculate $D \restriction \omega_1^M$.

Last we state a short lemma which will be helpful when showing that our coding forcings work the way they should.

Lemma 2.4. *Let T be a Suslin tree and let $\mathbb{A}_D(X)$ be the almost disjoint coding which codes a subset X of ω_1 into a real with the help of an almost disjoint family of reals D of size $\aleph_1$. Then*

$$\mathbb{A}_D(X) \Vdash T \textit{ is Suslin}$$

holds.

Proof. This is clear as $\mathbb{A}_D(X)$ has the Knaster property, thus the product $\mathbb{A}_D(X) \times T$ is ccc and T must be Suslin in $V^{\mathbb{A}_D(X)}$. $\square$

3 Coding machinery

We continue with the construction of the appropriate notions of forcing which we want to use in our proof. The goal is to first define a coding forcings $\mathrm{Code}(x)$ for reals x, which will force for x that a certain Σ^1_3-formula $\Phi(x)$ becomes true in the resulting generic extension. The coding method is basically the same as in [7] and [6].

In a first step we force over L to destroy all members of $\vec{S} = \vec{S}^0 \cup \vec{S}^1$ via generically adding an ω_1-branch, that is we form

$$\mathbb{P}^0 := \prod_{\alpha \in \omega_1} S_\alpha$$

with finite support. Note that this is an $\aleph_1$-sized, ccc forcing over L.

In a second step we add ω_1-many ω_1-Cohen subsets with a countably supported product, for technical reasons we want the second forcing to be defined as in L and not as in $L^{\mathbb{P}_0}$. We let

$$\mathbb{P}^1 := (\prod_{\alpha < \omega_1} \mathbb{C}(\omega_1))^L.$$

Note that this forcing is σ-closed only over L so we need to argue that the two step iteration $\mathbb{P}_0 * \mathbb{P}_1$ preserves $\aleph_1$. Forcing with the two step iteration $\mathbb{P}_0 * \mathbb{P}_1$ is the same as forcing with the product $\mathbb{P}_0 \times \mathbb{P}_1$ as the latter sits densely in the former. As $\mathbb{P}_0 \times \mathbb{P}_1$ is $\mathbb{P}_1 \times \mathbb{P}_0$ we can look at the latter and note that $\mathbb{P}_1$ is σ-closed over L, so in particular it does not add new ω_1-branches to trees from L [1]. Thus $\vec{T}$ remains an independent sequence of Suslin trees in $L^{\mathbb{P}_1}$ and $\mathbb{P}_0$ is a ccc forcing in $L^{\mathbb{P}_1}$. So $\aleph_1$ is preserved when forcing with $\mathbb{P}_0 \times \mathbb{P}_1$ and moreover and CH remains true.

We use W to denote this generic extension of L, that we fix a filter $G^0 \times G^1$ which is generic for $\mathbb{P}^0 \times \mathbb{P}^1$ and let

$$W = L[G^0][G^1].$$

Let $x \in W$ be a real, and let $m, k \in \omega$ and let $\eta < \omega_1$. We simply write (x, m, k) for a real w which codes the triple (x, m, k) in a recursive way. The forcing $\mathrm{Code}(x, m, k, 1, \eta)$ [2] which codes the triple (x, m, k) into $\vec{S^1}$ is defined

[1] Indeed any name of an ω_1-length branch $\dot{b}$ through some tree T gives rise to an ω_1 branch through T in L using the σ-closure of the forcing.

[2] The other coding forcing, $\mathrm{Code}(x, m, k, 0, \eta)$, which uses the trees from the $\vec{S}^0$-sequence instead is defined in the analogue way

as the almost disjoint coding forcing of a specific set $Y \subset \omega_1$, that is

$$\mathrm{Code}(x, m, k, 1) := \mathbb{A}(Y).$$

We will define the crucial set $Y \subset \omega_1$ now.

To ease notation we let $g \subset \omega_1$ be g_η for $\eta < \omega_1$, where g_η is the η-th coordinate of the $\prod_{\alpha<\omega_1} \mathbb{C}(\omega_1)$-generic filter over $L^{\mathbb{P}^0}$. We let $\rho : ([\omega_1]^\omega)^L \to \omega_1$ be some canonically definable, constructible bijection between these two sets. We use ρ and g to define the set $h \subset \omega_1$, which eventually shall be the set of indices of ω-blocks of $\vec{S}$, where we "code up the characteristic function of the real (x, m, k)", the latter slogan will be made precise in a moment. Let

$$h := \{\rho(g \cap \alpha) \,:\, \alpha < \omega_1\}$$

and let

$$\begin{aligned} A :=& \{\omega\gamma + 2n \mid \gamma \in h, n \notin (x, m, k)\} \cup \\ & \{\omega\gamma + 2n + 1 \mid \gamma \in h, n \in (x, m, k)\}. \end{aligned}$$

Let $X \subset \omega_1$ be chosen such that it canonically codes the following objects:

- The set $A \subset \omega_1$.
- The set $\{b_\beta \subset S^1_\beta \mid \beta \in A\}$ of generic ω_1-branches we added with $\mathbb{P}_0$.

Note that, when working in $L[X]$ and if $\gamma \in h$ then we can read off (x, m, k), and thus we say that (x, m, k) is coded into $\vec{S}^1$ at the ω-block starting at γ, via looking at the ω-block of $\vec{S}^1$-trees starting at γ and determine which tree has an ω_1-branch in $L[X]$.

$(*)(\gamma, (x, m, k))$ $n \in (x, m, k)$ if and only if $S^1_{\omega\cdot\gamma+2n+1}$ has an ω_1-branch, and $n \notin (x, m, k)$ if and only if $S^1_{\omega\cdot\gamma+2n}$ has an ω_1-branch.

Indeed if $n \notin (x, m, k)$ then we added a cofinal branch through $S^1_{\omega\cdot\gamma+2n}$. If on the other hand $S^1_{\omega\cdot\gamma+2n}$ does not have an ω_1-branch in $L[X]$ then we must have added an ω_1-branch through $S^1_{\omega\cdot\gamma+2n+1}$ as we always add an ω_1-branch through either $S^1_{\omega\cdot\gamma+2n+1}$ or $S^1_{\omega\cdot\gamma+2n}$ and adding branches through some S^1_α's will not affect that some S^1_β remain Suslin in $L[X]$, as $\vec{S}^1$ is independent.

We note that we can apply an argument resembling David's trick [3] in this situation. We rewrite the information of $X \subset \omega_1$ as a subset $Y \subset \omega_1$ using the following line of reasoning. Keeping lemma 2.2 in mind, it is clear that any transitive, $\aleph_1$-sized model N of ZF^- which contains X as an element will be able to first define $\vec{S}^1$ correctly and also correctly decode out of X all the

[3] see [2] for the original argument, where the strings in Jensen's coding machinery are altered such that certain unwanted universes are destroyed. This destruction is emulated in our context as seen below.

information regarding (x, m, k) being coded at each ω-block of $\vec{S}^1$ starting at every $\gamma \in h$. Consequently, if we code the model $(N, \in)$ as a set $X_N \subset \omega_1$, then for any uncountable β such that $L_\beta[X_N] \models \mathsf{ZF}^-$:

$$L_\beta[X_N] \models \text{“The model decoded out of } X_N \text{ satisfies } (*)_\gamma \text{ for every } \gamma \in h\text{”}.$$

For the latter assertion to be meaningful, we need that the model decoded out of X_N has h as an element but this is clear as h can be read off from A which in turn is coded into X which belongs to N as an element. We proceed by noting that there will be an $\aleph_1$-sized ordinal β as above and we can fix a club $C \subset \omega_1$ and a sequence $(M_\alpha : \alpha \in C)$ of countable elementary submodels of $L_\beta[X_N]$ such that

$$\forall \alpha \in C (M_\alpha \prec L_\beta[X_N] \wedge M_\alpha \cap \omega_1 = \alpha)$$

Now let the set $Y \subset \omega_1$ code the pair (C, X_N) such that the odd entries of Y should code X_N and if $E(Y)$ denotes the set of even entries of Y and $\{c_\alpha : \alpha < \omega_1\}$ is the enumeration of C then

1. $E(Y) \cap \omega$ codes a well-ordering of type c_0.
2. $E(Y) \cap [\omega, c_0) = \varnothing$.
3. For all β, $E(Y) \cap [c_\beta, c_\beta + \omega)$ codes a well-ordering of type $c_{\beta+1}$.
4. For all β, $E(Y) \cap [c_\beta + \omega, c_{\beta+1}) = \varnothing$.

We claim that the following assertion is true, which can be seen as a local version of $(*)(\gamma, (x, m, k))$.

$\sigma(x, m, k) :=$ For any countable transitive model M of “ZF^- and $\aleph_1$ exists” such that $\omega_1^M = (\omega_1^L)^M$ and $Y \cap \omega_1^M \in M$, M can construct its version of the universe $L[Y \cap \omega_1^M]$, and the latter will see that there is an $\aleph_1^M$-sized, transitive model $\bar{N} \in L[Y \cap \omega_1^M]$, $\bar{N} \models \mathsf{ZFC}^- \wedge \omega_1^M \in \bar{N}$ which models $(*)(\gamma, (x, m, k))$ for $\aleph_1^M$-many γ.

Indeed, if M is as asserted by $\sigma(x, m, k)$, then in particular $Y \cap \omega_1^M \in M$ and $\omega_1^M \in C$ by property 3 of Y. Thus $\omega_1^M = \omega_1^{M_\alpha}$ for an $M_\alpha \prec L_\beta[X_N]$ and the set which gets decoded out of $Y \cap \omega_1^M$ is the same no matter when we decode inside M_α or M. By elementarity, if M_α defines its version of $L[Y \cap \omega_1^M]$ internally then this inner model will see that there is a transitive $\bar{N} \models \mathsf{ZFC}^-$, $\omega_1^M \in \bar{N}$ which models $(*)(\gamma, (x, m, k))$ for $\aleph_1^M$-many γ. So by absoluteness of the decoding, M will find this $\bar{N}$ in its version of $L[Y \cap \omega_1^M]$ as well and again by absoluteness $\bar{N}$ satisfies the desired properties also when working in M, thus $\sigma(x, m, k)$ is true.

We have finally defined the desired set Y and now we use

$$\mathrm{Code}(x, m, k, 1, \eta) := \mathbb{A}(Y)$$

relative to our previously defined, almost disjoint family of reals $D \in L$ (see the paragraph after Definition 2.5) to code the set Y into one real r. This forcing only depends on the subset of ω_1 we code, thus $\mathbb{A}_D(Y)$ will be independent of the surrounding universe in which we define it, as long as it has the right ω_1 and contains the set Y.

The effect of the coding forcing $\mathrm{Code}(x, m, k, 1, \eta)$ is that it generically adds a real r such that this formula becomes true:

$\Psi(r, (x, m, k)) :=$ For any countable, transitive model M of "ZFC^- and $\aleph_1$ exists", such that $\omega_1^M = (\omega_1^L)^M$ and $r \in M$, M can construct its version of $L[r]$, denoted by $L[r]^M$, which in turn thinks that there is a transitive ZFC^--model $\bar{N}$ of size $\aleph_1^M$ such that $\bar{N}$ believes $(*)(\gamma, (x, m, k))$ for an $\aleph_1^M$-sized set of ordinals γ.

Indeed, if r and M are as above, then M and $L[r]^M$ will compute the almost disjoint family D up to the real indexed with $\omega_1 \cap M$ correctly, as discussed below the definition 2.3. As a consequence, $L[r]^M$ will contain the set $Y \cap \omega_1^M$, where $Y \subset \omega_1$ is as in the statement of $\sigma(x, m, k)$. So, as argued above, in $L[Y \cap \omega_1^M]$, there is an $\aleph_1^M$-sized, transitive $\bar{N}$ which models $(*)(\gamma, (x, m, k))$ for every $\gamma \in h \cap M$, as claimed.

Note that $\Psi(r, (x, m, k))$ is a Π^1_2-formula in the parameters r and (x, m, k), as the set $h \cap M \subset \omega_1^M$ is coded into r. We say in the above situation that the real (x, m, k) *is written into* $\vec{S}^1$, or that (x, m, k) *is coded into* $\vec{S}^1$. To summarize our discussion, given an arbitrary real of the form (x, m, k), then our forcing $\mathrm{Code}(x, m, k, 1)$, when applied over W, will add a real r which will turn the Π^1_2-formula $\Psi(r, (x, m, k))$ into a true statement in $W^{\mathrm{Code}(x,m,k,1)}$.

The coding forcing which codes a given real (x, m, k) into the $\vec{S}^0$, denoted by $\mathrm{Code}\,(x, m, k, 0, \eta)$ is defined in the same way.

The projective and local statement $\Psi(r, (x, m, k))$, if true, will determine how certain inner models of the surrounding universe will look like with respect to branches through $\vec{S}$. That is to say, if we assume that $\Psi(r, (x, m, k))$ holds. Then r also witnesses the truth of $\Psi(r, (x, m, k))$ for any transitive model M of the theory "$\mathsf{ZF}^- + \aleph_1$ exists and $\aleph_1 = \aleph_1^L$", which contains r (i.e. we can drop the assumption on the countability of M). Indeed if we assume that there would be an uncountable, transitive M, $r \in M$, which witnesses that $\Psi(r, (x, m, k))$ is false. Then by Löwenheim-Skolem, there would be a countable $N \prec M$, $r \in N$ which we can transitively collapse to obtain the transitive $\bar{N}$. But $\bar{N}$ would witness that $\Psi(r, (x, m, k))$ is not true for every countable, transitive model, which is a contradiction.

Consequently, the real r carries enough information that the universe $L[r]$ will see that certain trees from $\vec{S}^1$ have branches in that

$$n \in (x, m, k) \Rightarrow L[r] \models \text{“}S^1_{\omega\gamma+2n+1} \text{ has an } \omega_1\text{-branch”}.$$

and

$$n \notin (x, m, k) \Rightarrow L[r] \models \text{“}S^1_{\omega\gamma+2n} \text{ has an } \omega_1\text{-branch”}.$$

Indeed, the universe $L[r]$ will see that there is a transitive model N of “$\mathsf{ZF}^- + \aleph_1$ exists and $\aleph_1 = \aleph_1^L$” which believes $(*)$ for every $\gamma \in h \subset \omega_1$, the latter being coded into r. But by upwards Σ_1-absoluteness, and the fact that N can compute $\vec{S}^1$ correctly, if N thinks that some tree in $\vec{S}^1$ has a branch, then $L[r]$ must think so as well.

4 Allowable forcings

Next we define the set of forcings which we will use in our proof. We aim to iterate the coding forcings we defined in the last section.

Definition 4.1. *Let $W = L[G^0 \times G^1]$ be our ground model. Let $\alpha < \omega_1$ and let $F \in L$, $F : \alpha \to L$ be a bookkeeping function. A finite support iteration $\mathbb{P} = (\mathbb{P}_\beta : \beta < \alpha)$ is called allowable (relative to the bookkeeping function F) if the function $F : \alpha \to L$ determines $\mathbb{P}$ inductively as follows:*

> *We assume that $\beta \geqslant 0$ and $\mathbb{P}_\beta$ is defined. We let G_β be a $\mathbb{P}_\beta$-generic filter over W and assume that $F(\beta) = (\dot{x}, \dot{m}, \dot{k}, \dot{l}, \dot{\eta})$, for a quintuple of $(\mathbb{P}^0 \times \mathbb{P}^1) * \mathbb{P}_\beta$-names. We assume that $\dot{x}^{(G^0 \times G^1) * G_\beta} =: x$ is a real, $\dot{m}^{(G^0 \times G^1) * G_\beta} =: m$ and $\dot{k}^{(G^0 \times G^1) * G_\beta}$ are natural numbers, $\dot{l}^{(G^0 \times G^1) * G_\beta} \in \{0, 1\}$ and $\dot{\eta}^{(G^0 \times G^1) * G_\beta}$ is an ordinal $< \omega_1$.*
>
> *Then we split into two cases:*
>
> - *If there is a $\gamma < \beta$ and $F(\gamma) = (\dot{a}', \dot{m}', \dot{k}', \dot{l}', \dot{\eta}')$ for a quintuple of $\mathbb{P}_\gamma$-names $(\dot{a}', \dot{m}', \dot{k}', \dot{l}', \dot{\eta}')$ such that $\dot{a}'^{(G^0 \times G^1) * G_\gamma} = a \in \omega^\omega$, $\dot{m}'^{(G^0 \times G^1) * G_\gamma} = m' \in \omega$, $\dot{k}'^{(G^0 \times G^1) * G_\gamma} = k' \in \omega$, $\dot{\eta}'^{(G^0 \times G^1) * G_\gamma} = \eta$, in other words, if the η-th block where the iteration intends to code has already been used for coding at an earlier stage of the iteration, then we force with the trivial forcing.*
> - *If not, then let $\mathbb{P}(\beta)^{G_\beta} := \mathrm{Code}(x, m, k, l, \eta)$.*

If $\mathbb{P} \in W$ is a forcing such that there is an $\alpha < \omega_1$ and an $F \in L$, $F : \alpha \to L$ such that $\mathbb{P}$ is allowable with respect to F, then we often just drop the F and simply say that $\mathbb{P} \in W$ is allowable.

As allowable forcings form the base set of an inductively defined shrinking process, they are also denoted by 0-allowable with respect to F to emphasize

this fact. Informally speaking, the bookkeeping F hands us at every step a real of the form (x, m, k), decides whether to use $\vec{S}^0$ or $\vec{S}^1$ and uses a $\mathbb{C}(\omega_1)$-set (this set we will call a *coding area*) which gives rise to a subset of $h \subset \omega_1$ which corresponds to the places where we code up the relevant branches through $\vec{S}^l$ to compute (x, m, k) using the coding mechanism described in the previous section. The definition ensures that the sets h determined by the coding areas are an almost disjoint family of ω_1, i.e. two distinct such sets have countable intersection. The definition of allowable demands that each such coding area is used at most once in an allowable forcing, which will help to ensure that we will not accidentally code an unwanted (x, m, k) into $\vec{S}$.

Definition 4.2. *Let $\mathbb{P} = ((\mathbb{P}_\alpha, \mathbb{P}(\alpha)) \mid \alpha < \delta)$ be an allowable forcing. Let $G \subset \mathbb{P}$ be a generic filter over W. Then*

$$C^G := \{\eta < \omega_1 \mid \exists \beta < \delta \exists \dot{x}, \dot{m}, \dot{k}, \dot{l}, \dot{\eta} \in W^{\mathbb{P}_\beta} \\ (\mathbb{P}(\beta))^{G_\beta} = \mathrm{Code}(\dot{x}^{G_\beta}, \dot{m}^{G_\beta}, \dot{k}^{G_\beta}, \dot{l}^{G_\beta}, \dot{\eta}^{G_\beta} = \eta)\}$$

is the set of coding areas of $\mathbb{P}$ relative to G. We also let

$$C^{\mathbb{P}} := \{\eta < \omega_1 \mid \exists p \in \mathbb{P}(p \Vdash \eta \in C^{\dot{G}}\}.$$

It is immediate from the definition that C^G and also $C^{\mathbb{P}}$ are always countable sets for every allowable $\mathbb{P}$. Next we derive some properties of allowable forcings.

Lemma 4.3.

1. *If $\mathbb{P} = (\mathbb{P}(\beta) : \beta < \delta) \in W$ is allowable then for every $\beta < \delta$, $\mathbb{P}_\beta \Vdash |\mathbb{P}(\beta)| = \aleph_1$, thus every factor of $\mathbb{P}$ is forced to have size $\aleph_1$.*
2. *Every allowable forcing over W is ccc and thus preserves cardinals.*
3. *Every allowable forcing over W preserves* CH. *Furthermore, if $\mathbb{P} = (\mathbb{P}(\alpha) : \alpha < \omega_1) \in W$ is an ω_1-length iteration such that each initial segment of the iteration is allowable over W, then $W[\mathbb{P}] \models \mathsf{CH}$.*
4. *The product of two allowable forcings $\mathbb{P}$ and $\mathbb{Q}$ can be densely embedded into an allowable forcing provided that $C^{\mathbb{P}} \cap C^{\mathbb{Q}} = \varnothing$.*

Proof. The first, the second and the third assertion follow immediately from the definition modulo some well-known results.

The fourth item follows from the fact that almost disjoint coding forcings consist of finite conditions, thus their definition is absolute in every universe as long as the universe contains the subset of ω_1 the almost disjoint coding forcing wants to code. As a consequence the forcing $\mathbb{P} \times \mathbb{Q}$ embeds densely

into the finite support iteration of $\mathbb{P} * \check{\mathbb{Q}}$[4]. We claim that the iteration $\mathbb{P} * \check{\mathbb{Q}}$ is allowable. If $F_0 : \delta_0 \to L$ is the bookkeeping function which witnesses that $\mathbb{P}$ is allowable, and $F_1 : \delta_1 \to L$ witnesses that $\mathbb{Q}$ is allowable, then we define F with domain $\delta_0 + \delta_1$ as follows. We set $F(\beta) := F_0(\beta)$ if $\beta < \delta_0$ and $F(\delta_0 + \beta) := F_1(\beta)$. The bookkeeping F witnesses that $\mathbb{P} * \check{\mathbb{Q}}$ is allowable. This is trivial for $\beta < \delta_0$. And if we are at a stage $F(\delta_0 + \beta)$, then we will force with a coding forcing if and only if we use the same coding forcing when defining $\mathbb{Q}$ over W with the help of F_1 at stage β, which uses the fact that $C^{\mathbb{P}} \cap C^{\mathbb{Q}} = \varnothing$.

□

Every allowable forcing is determined by the associated F which is list of names of reals. We shall utilize this fact in showing that every allowable forcing can in fact be defined already in a proper inner model of W.

Lemma 4.4. *Let $\mathbb{P} \in W$ be an allowable forcing and let $F : \delta \to L$ be its bookkeeping. Then there is an uncountable, co-uncountable subset $I \subset \omega_1$ and a countable subset $J \subset \omega_1$ such that $\mathbb{P}$ can successfully be defined already in an inner model of W of the form $L[(G^0 \restriction I) \times (G^1 \restriction J)]$, where $G^0 \restriction I$ is just the restriction of the generic G^0 to coordinates which are in I and $G^1 \restriction J$ being defined similarly.*

Proof. The proof is via induction on the length δ of the iteration. Assume first that $\delta = 1$ then we can assume that $\mathbb{P} = \mathrm{Code}(x)$ for some real $x \in W$. Recall that W is defined as the generic extension of L via $\mathbb{P}^0$ which generically adds branches to each tree in $\vec{S}$ and $\mathbb{P}^1 = (\prod_{i\omega_1} \mathbb{C}(\omega_1))^L$. The reals in $W = L[G^0 \times G^1]$ are all elements of $L[G^0]$ already as is immediate from Easton's Lemma (see see Lemma 15.19 from [8]). So in particular there is a countable $I \subset \omega_1$ such that $w \in L[G^0 \restriction I]$ which shows the lemma for one step iterations.

Now we assume that the lemma is true for allowable forcings of length δ and our goal is to show that it also must be true for allowable forcings of length $\delta + 1$. We fix an allowable forcing $\mathbb{P}$ of length $\delta + 1$ and write $\mathbb{P} = \mathbb{P}_\delta * \mathrm{Code}(\dot{w})$ for some $(\mathbb{P}^0 \times \mathbb{P}^1) * \mathbb{P}_\delta$-name of a real $\dot{w}$. By our induction hypothesis $\mathbb{P}_\delta$ is already definable in some inner model $L[G^0 \restriction I][G^1 \restriction J]$. So the name $\dot{w}$ can be written as a $[(\mathbb{P}^0 \restriction I) \times (\mathbb{P}^1 \restriction J) * \mathbb{P}_\delta] \times [(\mathbb{P}^0 \restriction (\omega_1 \backslash I)) \times \mathbb{P}^1 \restriction (\omega_1 \backslash J)]$-name. Again by Easton's Lemma, this time applied over the ground model $L[G^1 \restriction J]$, we obtain that the name $\dot{w}$ is in fact

[4] Let $i : \mathbb{P} \times \mathbb{Q} \to \mathbb{P} * \check{\mathbb{Q}}$ be the canonical embedding defined by concatenation. To show that i has a dense image, fix an arbitrary condition $r \in \mathbb{P} * \check{\mathbb{Q}}$. Because the iteration $\mathbb{P} * \check{\mathbb{Q}}$ uses finite support, the tail $r \restriction [\delta_1, \delta_1 + \beta)$ has only finitely many non-trivial coordinates. We can iteratively extend the head $r \restriction \alpha \in \mathbb{P}$ finitely many times to obtain a condition $p_1 \leqslant r \restriction \alpha$ in $\mathbb{P}$ that completely decides the ground-model values of the tail at these coordinates. These decided values naturally define a condition $p_2 \in \mathbb{Q}$ (which has finite support since r does), yielding a ground-model pair (p_1, p_2) such that $i(p_1, p_2) \leqslant r$.

(equivalent to) a $[(\mathbb{P}^0 \restriction I) \times (\mathbb{P}^1 \restriction J) * \mathbb{P}_\delta] \times (\mathbb{P}^0 \restriction (\omega_1 \backslash I))$-name. Note that the forcing $\mathbb{P}^0 \restriction (\omega_1 \backslash I)$ is an ω_1-length iteration with finite support which adds a branch through every Suslin tree indexed in the set $\omega_1 \backslash I$ so by standard facts of forcing theory the name $\dot{w}$ is in fact (equivalent to) a $(\mathbb{P}^0 \restriction I) \times (\mathbb{P}^1 \restriction J) * \mathbb{P}_\delta \times (\mathbb{P}^0 \restriction \tilde{I}))$-name, for a countable set $\tilde{I}$. We add $\tilde{I}$ to I and finally note that the coding forcing $\mathrm{Code}(\dot{w})$, as it is ccc will potentially only use a countable set of coding areas $\tilde{J}$. So $\mathbb{P} = \mathbb{P}_\delta * \mathrm{Code}(\dot{w})$ can successfully be defined in the inner model $L[G^0 \restriction I \cup \tilde{I}][G^1 \restriction J \cup \tilde{J}]$ which shows the successor case.

The limit case follows immediately from the above via taking countable unions. □

Let $\mathbb{P} = (\mathbb{P}(\beta) : \beta < \delta)$ be an allowable forcing with respect to some $F \in W$. The set of (names of) reals which are enumerated by F, and where the second case in the definition of allowable applies, i.e. we actually used a coding forcing, is dubbed *he set of reals which are coded by* $\mathbb{P}$. That is, for every β, if we let $\dot{x}_\beta$ be the (name) of a real listed by $F(\beta)$ and if we let $G \subset \mathbb{P}$ be a generic filter over W and finally if we let $\dot{x}_\beta^G =: x_\beta$, then we say that $\{x_\beta : \beta < \delta\}$ is the set of reals coded by $\mathbb{P}$ and G (though we will suppress the G).

Next we show, that iterations of 0-allowable forcings will not add accidentally new elements to the set of reals defined by the Σ^1_3-formula

$$\Phi_1((x, m, k)) := \exists r \Psi_1(r, (x, m, k)),$$

where Ψ_1 is defined in the previous section and by the same argument will also not add non-intended members to the set defined by

$$\Phi_0 := \exists r \Psi_0(r, (x, m, k)).$$

We let

$$\Phi(x) \equiv \Phi_0(x) \vee \Phi_1(x).$$

To utilize already established jargon we will say that "x is coded into $\vec{S}^i$" whenever $\Phi_i(x)$ is true. Likewise, if $\neg\Phi_i(x)$ holds true, then we say that "x is not coded into $\vec{S}^i$".

Lemma 4.5. *If $\mathbb{P} \in W$ is allowable, $\mathbb{P} = (\mathbb{P}_\beta : \beta < \delta)$, $G \subset \mathbb{P}$ is generic over W and $\{x_\beta : \beta < \delta\}$ is the set of reals which are coded by $\mathbb{P}$. Let $\Phi(v_0)$ be the distinguished formula from above. Then in $W[G]$, the set of reals which satisfy $\Phi(v_0)$ is exactly $\{x_\beta : \beta < \delta\}$.*

Proof. Let G be $\mathbb{P}$ generic over W, we work in $W[G]$. Let $g = (g_\beta : \beta < \delta)$ be the set of the δ-many coding areas of $\mathbb{P}$ relative to G. We let $\rho : ([\omega_1]^\omega)^L \to \omega_1$ be our fixed, constructible bijection and let $h_\beta = \{\rho(g_\beta \cap \alpha) : \alpha < \delta\}$. Note that the family $\{h_\beta : \beta < \delta\}$ forms an almost disjoint family of subsets

of ω_1, that is two distinct members of $\{h_\beta : \beta < \delta\}$ always have countable intersection. Thus there is an $\eta < \omega_1$ such that for arbitrary distinct β_1, $\beta_2 < \delta$, $h_{\beta_1} \cap h_{\beta_2} \subset \eta$.

We assume for a contradiction, that there is a real (x, m, k) which satisfies $\Psi_1((x, m, k))$ but $(x, m, k) \notin \{x_\beta \mid \beta < \delta\}$. As a consequence, by the discussion right at the end of the third section, r codes an unbounded set $h \subset \omega_1$ such that for every $\gamma \in h$ the following holds true.

$$n \in (x, m, k) \Rightarrow L[r] \models \text{“}S^1_{\omega\gamma+2n+1} \text{ has an } \omega_1\text{-branch”}.$$

and

$$n \notin (x, m, k) \Rightarrow L[r] \models \text{“}S^1_{\omega\gamma+2n} \text{ has an } \omega_1\text{-branch”}.$$

As (x, m, k) is distinct from every x_β, there must be $\aleph_1$-many $\alpha > \eta$ and an $n \in \omega$ such that, without loss of generality,

$$L[r] \models \text{“}S^1_{\omega\alpha+2n+1} \text{ has an } \omega_1\text{-branch”},$$

yet for every r_β witnessing that $\Psi_1(r_\beta, x_\beta)$ holds true

$$L[r_\beta] \models \text{“}S^1_{\omega\alpha+2n+1} \text{ does not have an } \omega_1\text{-branch”}.$$

Invoking the (proof of the) last lemma there is an uncountable, co-uncountable $I \subset \omega_1$ and a countable $J \subset \omega_1$ such that $\mathbb{P} \in L[G^0 \restriction I][G^1 \restriction J]$ and the set I consists of all the indices of trees determined by the coding areas $G^1 \restriction J$ with the addition of only countably many indices from $\vec{S}$. We denote this countable set of indices with C. We collect the sets

$$\begin{aligned} A_\beta :=& \{\omega\gamma + 2n \mid \gamma \in h_\beta, n \notin x_\beta\} \cup \\ & \{\omega\gamma + 2n + 1 \mid \gamma \in h_\beta, n \in x_\beta\}. \end{aligned}$$

As $G \subset \mathbb{P}$ picks at each stage $\beta < \delta$ of its iteration exactly one coding area g_β, the generic extension

$$L[G^0 \restriction I][G^1 \restriction J][G]$$

can be re-written as

$$L[\{G^0_\xi \mid \xi \in \bigcup_{\beta<\delta} A_\beta\}][\{G^0_\xi \mid \xi \in C\}][(g_\beta \mid \beta < \delta)][\{G^0_\xi \mid \xi \in I \wedge \xi \notin (C \cup \bigcup_{\beta<\delta} A_\beta)\}].$$

In particular each tree from $\vec{S}$ with index not in $C \cup \bigcup_{\beta<\delta} A_\beta$ is still Suslin in the inner model $L[\{G^0_\xi \mid \exists\beta < \delta(\xi \in A_\beta\}][\{G^i_\xi \mid \xi \in C\}][(g_\beta \mid \beta < \delta)]$ by the independence of the sequence.

It is therefore possible to fix an index $\alpha \notin C \cup \bigcup_{\beta<\delta} A_\beta$ such that there is a real r with

$$L[r] \models \text{“}S^1_{\omega\alpha+2n+1} \text{ has an } \omega_1\text{-branch”}$$

whereas for every $\beta < \delta$

$$L[r_\beta] \models \text{“}S^1_{\omega\alpha+2n+1} \text{ does not have an } \omega_1\text{-branch”}.$$

We claim however that there is no real in $W[G]$ such that $W[G] \models L[r] \models$ “$S^1_{\omega\alpha+2n+1}$ has an ω_1-branch”, which will be the desired contradiction.

We show this by pulling out the forcing $S^1_{\omega\alpha+2n+1}$ out of the forcing $\mathbb{P}^0 \times \mathbb{P}^1 * \mathbb{P}$ over L which produces $W[G]$. Indeed if we consider $W[\mathbb{P}] = L[\mathbb{P}^0][\mathbb{P}^1][\mathbb{P}]$, and if $S^1_{\omega\alpha+2n+1}$ is as described already, by the reasoning above we can rearrange the generics to $W[G] = L[G'^0 \times G^0_{\omega\alpha+2n+1}][G^1][G] = L[G'][G^0_{\omega\alpha+2n+1}]$, where G'^0 is $\prod_{\beta\neq\alpha+2n+1} G^0_\beta$ and G' is $G'^0 \times G^1 * G$.

Note now that $S_{\alpha+2n+1}$ is still a Suslin tree in $L[G']$ so the forcing $S^1_{\alpha+2n+1}$ is ω-distributive. This can be seen using the fact that $\vec{S}^0$ and $\vec{S}^1$ are independent. Indeed $S^1_{\alpha+2n+1}$ will remain Suslin in $L[G'^0][G^1]$, as we can write the universe as $L[G^1][G'^0]$. But then the finite support iteration of the almost disjoint coding forcings is Knaster, hence keeps $S^1_{\alpha+2n+1}$ Suslin. Consequently $2^\omega \cap W[G] = 2^\omega \cap L[G']$.

But this implies that

$$L[G'] \models \neg\exists r L[r] \models \text{“}S^1_{\alpha+2n+1} \text{ has an } \omega_1\text{-branch”}$$

as the existence of an ω_1-branch through $S^1_{\alpha+2n+1}$ in the inner model $L[r]$ would imply the existence of such a branch in $L[G']$. Further, as no new reals appear when passing to $W[G]$ we also get

$$W[G] \models \neg\exists r L[r] \models \text{“}S^1_{\alpha+2n+1} \text{ has an } \omega_1\text{-branch”}.$$

This is the desired contradiction.

□

4.1 α-allowable forcings

The notion of 0-allowable will form the base case of an inductive definition. Let $\alpha \geqslant 0$ be an ordinal and assume we defined already the notion of α-allowable relative to a bookkeeping F and a set E. For 0-allowable we define the set E to be empty. Then we can inductively define the notion of $\alpha + 1$-allowable with respect to E and F as follows.

We list all Σ^1_3-formulas $\varphi_n(v_0, v_1)$ with two free variables. If y is a real then $\varphi_n(v_0, y)$ defines a $\Sigma^1_3(y)$-set $A_n(y_0)$, where $x \in A_n(y) \leftrightarrow \varphi_n(x, y)$. Suppose that $\gamma < \omega_1$, F is a bookkeeping function,

$$F : \gamma \to H(\omega_2)$$

and

$$\mathbb{P} = (\mathbb{P}_\beta \,:\, \beta < \gamma)$$

is a allowable forcing relative to F.

Suppose that

$$E = E_0 \cup E_1$$

where

$$E_0 := \{(\dot{y}_\delta, \dot{m}_\delta, \dot{k}_\delta) \,:\, \delta \leqslant \alpha\}$$

and

$$E_1 := \{(\dot{x}_\delta, i_\delta) \,:\, \delta \leqslant \alpha, i_\delta \in \{0,1\}\}$$

where we allow $(\dot{y}_\delta, \dot{m}_\delta, \dot{k}_\delta)$ and $(\dot{x}_\delta, i_\delta)$ to be the empty set. Otherwise $\dot{m}_\delta, \dot{k}_\delta$ are names for elements of ω and $\dot{x}_\delta, \dot{y}_\delta$ are $\mathbb{P}$-names of two reals and for every two ordinals $\beta < \gamma \leqslant \alpha$, if $(\dot{y}_\beta, \dot{m}_\delta, \dot{k}_\delta)$ and $(\dot{y}_\gamma, \dot{m}_\gamma, \dot{k}_\gamma)$ are not the empty set, then $\mathbb{P} \Vdash (\dot{y}_\beta, \dot{m}_\beta, k_\beta) \neq (\dot{y}_\gamma, \dot{m}_\gamma, k_\gamma)$. Intuitively, E_0 will serve as the set of pairs of boldface Σ^1_3-sets, for which we already obtained rules which allow us to separate them; whereas E_1 is the set of (names of) reals which we decided to never code along our α-allowable iteration using the coding forcing $\mathrm{Code}(x, a, b, i, \eta)$ for two fixed natural numbers a and b and any ordinal $\eta < \omega_1$. The latter plays a key role in establishing the eventual failure of Π^1_3-reduction.

Suppose that for every $\delta \leqslant \alpha$, $(\mathbb{P}_\beta \,:\, \beta < \gamma)$ is δ-allowable with respect to $E \restriction \delta = (E_0 \restriction \delta) \cup (E_1 \restriction \delta) = \{(\dot{y}_\eta, \dot{m}_\eta, k_\eta) \,:\, \eta < \delta\} \cup \{(\dot{x}_\eta, i_\eta) \,:\, \eta < \delta\}$ and F. Our definition will consider two cases. In the first case we shall define $\alpha+1$-allowable with respect to $E_0 \cup \{\dot{y}_{\alpha+1}, \dot{m}_{\alpha+1}, \dot{k}_{\alpha+1})\}$, that is we enlarge E_0 while keeping the old E_1 (via letting $(\dot{x}_{\alpha+1}, i_{\alpha+1})$ be the empty set) and say what $\alpha+1$-allowable should mean. In the second case we enlarge E_1, while keeping the old E_0 and say what $\alpha+1$-allowable should mean. We assume as a part of the definition of α-allowable that we did not enlarge both E_0 and E_1 when passing from β-allowable to $\beta+1$-allowable for any $\beta < \alpha$. This property will be kept when defining $\alpha+1$-allowable which is what we do now.

4.1.1 $\alpha+1$-allowable forcings

Suppose $\mathbb{P} = (\mathbb{P}_\beta \,:\, \beta < \gamma)$ is α-allowable with respect to $E = E_0 \cup E_1$ and a bookkeeping function F. We define the notion of $\alpha+1$-allowable by specifying two mutually exclusive ways to extend E: either enlarging E_0 or enlarging E_1.

The Freshness Condition: Throughout the rules below, we require a recurring check to ensure a coding area hasn't been used at an earlier stage. Let us define the **Freshness Condition** at stage β for a given η as follows:

> There is no $\gamma < \beta$ and no $\mathbb{P}_\gamma$-name of a tuple $(\dot{a}, \dot{m}', \dot{k}', \dot{l}', \dot{\eta})$ such that $\dot{a}^{G_\gamma} = a \in \omega^\omega$, $\dot{m}'^{G_\gamma} = m' \in \omega$, $\dot{k}'^{G_\gamma} = k' \in \omega$, $\dot{l}'^{G_\gamma} \in \{0,1\}$, $\dot{\eta}^{G_\gamma} = \eta$, and $\mathbb{P}(\gamma)^{G_\gamma} = Code(a, m', k', l', \eta)$.

If the Freshness Condition **fails** at stage β, we always force with the trivial forcing.

Case 1: Enlarging E_0 Assume $(\dot{x}_{\alpha+1}, i_{\alpha+1})$ is empty, $\dot{y}_{\alpha+1}$ is a $\mathbb{P}$-name for a real, and $m_{\alpha+1}, k_{\alpha+1} \in \omega$ such that $\mathbb{P} \Vdash \forall \delta \leqslant \alpha\, ((\dot{y}_\delta, m_\delta, k_\delta) \neq (\dot{y}_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1}))$. We define $\mathbb{P}$ to be $\alpha+1$-allowable with respect to F and $E' = E_0 \cup \{(\dot{y}_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1})\} \cup E_1$ if it obeys the following rules for every $\beta < \gamma$. Let G_β be $\mathbb{P}_\beta$-generic over W, and evaluate the names given by $F(\beta)$ using G_β:

1. **Positive Separation (A_m):** If $F(\beta) = (\dot{x}, \dot{y}_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1}, i, \dot{\eta})$ and $W[G_\beta]$ satisfies:

   $$\exists \mathbb{Q}\, (\mathbb{Q} \text{ is 0-allowable w.r.t } E \text{ and some } F' \wedge \mathbb{Q} \Vdash x \in A_m(y_{\alpha+1}))$$

   Then, provided the Freshness Condition holds, the following equation must be true:

   $$\mathbb{P}(\beta)^{G_\beta} := Code(x, y_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1}, 0, \eta)$$

2. **Negative Separation (A_k):** If Rule 1 fails, $F(\beta) = (\dot{x}, \dot{y}_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1}, i, \dot{\eta})$, and $W[G_\beta]$ satisfies:

   $$\exists \mathbb{Q}_2\, (\mathbb{Q}_2 \text{ is 0-allowable w.r.t } E \text{ and some } F' \wedge \mathbb{Q}_2 \Vdash x \in A_k(y_{\alpha+1}))$$

   Then, provided the Freshness Condition holds, the following must be true:

   $$\mathbb{P}(\beta)^{G_\beta} := Code(x, y_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1}, 1, \eta)$$

3. **Default Coding (Target variables):** If neither Rule 1 nor Rule 2 applies for $F(\beta) = (\dot{x}, \dot{y}_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1}, i, \dot{\eta})$, and the Freshness Condition holds, the following must be true:

   $$\mathbb{P}(\beta)^{G_\beta} := Code(x, y_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1}, i^{G_\beta}, \eta)$$

4. **Default Coding (Other variables):** If $F(\beta) = (\dot{x}, \dot{y}, \dot{m}, k, i, \dot{\eta})$ evaluates to a tuple such that:

   $$W[G_\beta] \models ((y, m, k) \notin E^{G_\beta} \cup \{(y_{\alpha+1}, m_{\alpha+1}, k_{\alpha+1})\})$$

   where E^{G_β} is the set of evaluations of names in E with the help of G_β. Then, provided the Freshness Condition holds, the following must be true:

   $$\mathbb{P}(\beta)^{G_\beta} := Code(x, y, m, k, i, \eta)$$

Case 2: Enlarging E_1 Assume $(\dot{y}_{\alpha+1}, \dot{m}_{\alpha+1}, \dot{k}_{\alpha+1})$ is empty, $\dot{x}_{\alpha+1}$ is a $\mathbb{P}$-name of a real, and $i_{\alpha+1}$ is a $\mathbb{P}$-name of an element in $\{0,1\}$.

We define $\mathbb{P}$ to be $\alpha+1$-allowable with respect to F and $E' = E_0 \cup E_1 \cup \{(\dot{x}_{\alpha+1}, i_{\alpha+1})\}$ if it is α-allowable relative to E and F with the following additional constraint:

> For any $\eta < \omega_1$, the forcing $Code(\dot{x}_{\alpha+1}^{G_\beta}, a, b, i_{\alpha+1}^{G_\beta}, \eta)$ is never used as a factor $\mathbb{P}(\beta)^{G_\beta}$, provided both $\dot{x}_{\alpha+1}$ and $i_{\alpha+1}$ are $\mathbb{P}_\beta$-names that can be evaluated with G_β. (Here a, b represent the Gödel numbers of our two distinguished Σ^1_3 formulas).

Formally, to never use this forcing as a factor means that for our iteration $\mathbb{P} = (\mathbb{P}_\beta \mid \beta < \gamma)$:

$$\forall \beta < \gamma \Big((\dot{x}_{\alpha+1}, i_{\alpha+1} \text{ are } \mathbb{P}_\beta\text{-names}) \implies \forall \eta < \omega_1 \big(\mathbb{P}(\beta)^{G_\beta} \neq Code(\dot{x}_{\alpha+1}^{G_\beta}, a, b, i_{\alpha+1}^{G_\beta}, \eta) \big) \Big).$$

We also allow enlarging E_1 by any finite set of pairs (of names for reals and indicators in $\{0,1\}$), with the demand that we never use the associated coding forcings when defining $\alpha+1$-allowable iterations with respect to E' and F.

4.1.2 Limit stages

For limit ordinals α, we say that an allowable forcing $\mathbb{P}$ is α allowable with respect to E and F if for every $\eta < \alpha$, $(\mathbb{P}_\beta : \beta < \gamma)$ is η-allowable with respect to $E \restriction \eta$ and some F'.

4.1.3 Remarks

We add a couple of remarks concerning the last definition.

- By definition, if $\delta_2 < \delta_1$ and $\mathbb{P}_1$ is δ_1-allowable with respect to $E = \{(\dot{y}_\beta, m_\beta, k_\beta) : \beta \leqslant \delta_1\} \cup \{(\dot{x}_\beta, i_\beta) : \beta \leqslant \delta_1$ and some F_1, then $\mathbb{P}_1$ is also δ_2-allowable with respect to $E \restriction \delta_2 = \{(\dot{y}_\beta, m_\beta, k_\beta) : \beta \leqslant \delta_2\} \cup \{(\dot{x}_\beta, i_\beta) : \beta \leqslant \delta_2, i_\beta \in \{0,1\}\}$ and an altered bookkeeping function F'. The bookkeeping F' should just use the relevant stages where case 1.3 applies to guess what the δ_1-allowable forcing $\mathbb{P}_1$ with respect to E and F does at these stages when in cases 1.1 or 1.2.

- The notion of α-allowable can be defined in a uniform way over any allowable extension W' of W.

- We will often just say that some iteration $\mathbb{P}$ is α-allowable, by which we mean that there is a set E and a bookkeeping F such that $\mathbb{P}$ is α-allowable with respect to E and F.

5 Closure under products

Lemma 5.1. *Let α be an ordinal, assume that W' is some α-allowable generic extension of W, and that $\mathbb{P}^1 = (\mathbb{P}^1_\beta : \beta < \delta_1)$ and $\mathbb{P}^2 = (\mathbb{P}^2_\beta : \beta < \delta_2)$ are two α-allowable forcings over W' with respect to a common set $E = E_0 \cup E_1 = \{(\dot{y}_\delta, \dot{m}_\delta, \dot{k}_\delta) : \delta < \alpha\} \cup \{(\dot{x}_\delta, i_\delta) : \delta < \alpha\}$ and bookkeeping functions F_1 and F_2 respectively. Further assume that*

$$C(\mathbb{P}^1) \cap C(\mathbb{P}^2) = \varnothing,$$

and that $E_1 = \{(\dot{x}_\delta, i_\delta) \mid \delta < \alpha\}$ consists of reals which are in fact elements of W'. Then there is a bookkeeping function F such that $\mathbb{P}^1 \times \mathbb{P}^2$ densely embeds into an α-allowable forcing over W' with respect to E and F.

Proof. We shall show that $\mathbb{P}^1 * \check{\mathbb{P}}^2$ is α-allowable with respect to E and some F. This iteration should then serve as the asserted α-allowable forcing in which $\mathbb{P}^1 \times \mathbb{P}^2$ densely embeds. We define $F \restriction \delta_1$ to be F_1. For values $\delta_1 + \beta \geqslant \delta_1$ we let $F(\delta_1 + \beta)$ be $F_2(\beta)$ where we identify $\mathbb{P}^2_\beta$-names with the $\mathbb{P}^1$-check names for $\mathbb{P}^2_\beta$-names. The lemma is proved by induction on α. If $\alpha = 0$, then this is Lemma 4.3.

So we assume that $\alpha = 1$. We assume first that, we have enlarged E_1 in order to get the notion of 1-allowable. Then there is a real $x_0 \in W'$ and $i \in \{0, 1\}$ such that 1-allowable with respect to E and some F is just an allowable forcing with the additional constraint, that we must not use forcings of the form $\mathrm{Code}(x_0, y, a, b, i, \eta)$ for every η. This means that

$$\forall \eta < \delta_1 (\forall \zeta < \omega_1 ((\mathbb{P}^1(\eta)^{G^1_\eta} \neq Code(x_0, a, b, i, \zeta))).$$

and

$$\forall \eta < \delta_2 (\forall \zeta < \omega_1 ((\mathbb{P}^2(\eta)^{G^2_\eta} \neq Code(x_0, a, b, i, \zeta))).$$

This property is clearly preserved under $\mathbb{P}^1 \times \mathbb{P}^2$. So if both $\mathbb{P}^1$ and $\mathbb{P}^2_\beta$ are 1-allowable with respect to E and F_1 and F_2 respectively, then, at stage β $\mathbb{P}^1 \times \mathbb{P}^2$ is 1-allowable with respect to F and E as asserted.

So we can assume that 1-allowable was obtained via enlarging $E = E_0 = \{(\dot{y}, \dot{m}, \dot{k})\}$. Let $(\mathbb{P}_2)_\beta$ be the iteration of $\mathbb{P}_2$ up to stage $\beta < \delta_2$. Assume, that $\mathbb{P}_1 \times (\mathbb{P}_2)_\beta$ is in fact an α-allowable forcing relative to E and F. Let $(G^1 \times G^2_\beta)$ be a $\mathbb{P}^1 \times \mathbb{P}^2_\beta$-generic filter over W'. Then we have that $F(\delta_1 + \beta) = F_2(\beta) = (\dot{x}, \dot{y}, \dot{m}, \dot{k}, i, \dot{\eta})$, and we claim that

Claim 1. *At any stage $\beta < \delta_2$, if we evaluate the names in $F_2(\beta)$ using G^2_β, the specific case (Case 1.1, 1.2, 1.3, 1.4 from the definition of α-allowability) that applies in the model $W'[G^2_\beta]$ is exactly the same case that applies in the extended model $W'[G^1][G^2_\beta]$.*

If the claim holds, the lemma follows immediately by induction on $\beta < \delta_2$. Working in $W'[G^1][G^2_\beta]$, we evaluate $F(\delta_1 + \beta) = F_2(\beta)$. By the claim, the case determining $\mathbb{P}^2(\beta)^{G^2_\beta}$ over the larger model $W'[G^1][G^2_\beta]$ is identical to the one over $W'[G^2_\beta]$, ensuring the rules of α-allowability are strictly obeyed.

We assume first that at stage β, in the definition of $\mathbb{P}^2$, case 1.1 in the definition of 1-allowable applies, when working in the model $W'[G^2_\beta]$ relative to E and F_2. So we assume that β is a stage such that

$$F_2(\beta) = (\dot{x}, \dot{y}, \dot{m}, \dot{k}, i, \dot{\eta})$$

and $(\dot{y}, \dot{m}, \dot{k}) \in E$ and if $\dot{x}^{G^2_\beta} = x$ and $\dot{y}^{G^2_\beta} = y$, the universe $W'[G^2_\beta]$ thinks that

$$\begin{aligned}&\exists \mathbb{Q}(\mathbb{Q} \text{ is 0-allowable with respect to } E \text{ and some F'} \wedge \\ &\quad \mathbb{Q} \Vdash x \in A_m(y)).\end{aligned}$$

Now when working over $W'[G^1 \times G^2_\beta]$ instead, we shall show that we are still in case 1.1.

The idea is that $\mathbb{Q}$, as it will force $x \in A_m(y)$, will witness that we are in case 1 as well when working over $W[G^1][G^2_\beta]$. For that we need to show that $\mathbb{P}^1 \times \mathbb{Q}$ is 0-allowable, or in other words, that their coding areas do not intersect. As there is no reason why this should be the case, we need an additional argument. For that we show that the existence of a 0-allowable forcing which forces some Σ^1_3-statement is not tied to the coding areas it uses, as they are generically added sets. This will imply that we can shift the coding areas of $\mathbb{Q}$ away from C^{G^1} and still obtain allowable forcings using these shifted coding areas instead while still forcing the Σ^1_3-assertion to hold.

We shall explain this shifting mechanism now. We write $W = L[G]$ and let $(G)_0$ and $(G)_1$ denote its projection to the first and the second coordinate respectively, so that $(G)_0$ is $\mathbb{P}^0 = \prod \vec{S}$-generic and $(G)_1$ is $\mathbb{P}^1 = \prod_{i<\omega_1} \mathbb{C}(\omega_1)^L$-generic. We write $(G)_1$ as $(C_i \mid i < \omega_1)$. As G^2_β picks at each stage exactly one coding area C_i there is $\eta < \omega_1$ and we can write $L[(G)_0][\{C_i \mid i < \omega_1\}][G^2_\beta]$ as $L[(G)_0][\{C_i \mid i < \eta\}][G^2_\beta][\{C_i \mid \eta \leqslant i < \omega_1\}]$ and the coding areas of $\mathbb{P}^2_\beta$ relative to G^2_β is a subset of η.

$$\begin{aligned}L[(G)_0][\{C_i \mid i < \eta\}][G^2_\beta][\{C_{\eta+j} \mid j < \omega_1\}] \models \exists \mathbb{Q}(&\mathbb{Q} \text{ is 0-allowable with} \\ &\text{respect to } E \text{ and some F'} \\ &\wedge \mathbb{Q} \Vdash x \in A_m(y)).\end{aligned}$$

Thus there is a condition $p \in \prod_{j<\omega_1} C_{\eta+j} \subset (\prod_{\eta \leqslant j < \omega_1} \mathbb{C}(\omega_1))^L$ such that

$$\begin{aligned}L[G][\{C_i \mid i < \eta\}][G^2_\beta] \models p \Vdash_{(\prod_{j\in[\eta,\omega_1)} \mathbb{C}(\omega_1))^L} \exists \mathbb{Q}(&\mathbb{Q} \text{ is 0-allowable with} \\ &\text{respect to } E \text{ and some F'} \\ &\wedge \mathbb{Q} \Vdash x \in A_m(y)).\end{aligned}$$

As $(\prod_{j\in[\eta,\omega_1)}\mathbb{C}(\omega_1))^L$ is isomorphic to any $(\prod_{i\in[\zeta,\omega_1)}\mathbb{C}(\omega_1))^L$, for $\zeta<\omega_1$, we can find a sufficiently large $\zeta<\omega_1$ such that the interval $[\zeta,\omega_1)$ is fully disjoint from the coding areas from $\mathbb{P}^1$ relative to G^1. We can also find a condition $p'\in(\prod_{j\in[\zeta,\omega_1)}\mathbb{C}(\omega_1))^L$ such that

$$L[G][\{C_i \mid i<\eta\}][G^2_\beta]\models p' \Vdash_{(\prod_{j\in[\zeta,\omega_1)}\mathbb{C}(\omega_1))^L} \exists\mathbb{Q}(\mathbb{Q} \text{ is 0-allowable with respect to } E \text{ and some } F' \wedge \mathbb{Q}\Vdash x\in A_m(y)).$$

Note that $(\prod_{i<\omega_1}\mathbb{C}(\omega_1))^L$ is weakly homogeneous [5], so in fact

$$L[G][\{C_i \mid i<\eta\}][G^2_\beta]\models 1 \Vdash_{(\prod_{j\in[\zeta,\omega_1)}\mathbb{C}(\omega_1))^L} \exists\mathbb{Q}(\mathbb{Q} \text{ is 0-allowable with respect to } E \text{ and some } F' \wedge \mathbb{Q}\Vdash x\in A_m(y)).$$

So for any $\zeta<\omega_1$, $\zeta>\eta$ we can find a condition $q\in(\prod_{j\in[\zeta,\omega_1)}\mathbb{C}(\omega_1))^L$ such that $q\in\prod_{j\in[\zeta,\omega_1)}C_j$ and

$$L[G][\{C_i \mid i<\eta\}][G^2_\beta]\models q \Vdash_{(\prod_{j\in[\zeta,\omega_1)}\mathbb{C}(\omega_1))^L} \exists\mathbb{Q}(\mathbb{Q} \text{ is 0-allowable with respect to } E \text{ and some } F' \wedge \mathbb{Q}\Vdash x\in A_m(y)).$$

As the coding areas of $\mathbb{P}^1$ are countable, we can find a $\zeta>\eta$ such that $C^{\mathbb{P}^1}\cap[\zeta,\omega_1)=\varnothing$. Also the coding areas of $\mathbb{Q}$ are contained as a subset in the interval $[\zeta,\omega_1)$ so we can conclude

$$L[(G)_0][(G)_1][G^2_\beta]\models \exists\mathbb{Q}(\mathbb{Q} \text{ is 0-allowable with respect to } E \text{ and some } F' \wedge C^{\mathbb{Q}}\cap C^{\mathbb{P}^1}=\varnothing \wedge \mathbb{Q}\Vdash x\in A_m(y)).$$

and so

$$L[(G)_0][(G)_1][G^1][G^2_\beta]\models \exists\mathbb{Q}(\mathbb{Q} \text{ is 0-allowable with respect to } E \text{ and some } F' \wedge \mathbb{Q}\Vdash x\in A_m(y)).$$

So at stage β, we are in case 1 as well, when working over $W[G^1][G^2_\beta]$ which proves the Claim in the first instance.

[5] The homogeneity of the forcing is in fact not necessarily needed for this argument and its use can be replaced by a density argument. We use the homogeneity as the argument becomes neater this way.

If, at stage β and working in the model $W[G^2_\beta]$ case 2 applies, when considering $\mathbb{P}_2$ as a 1-allowable forcing with respect to E and F^2 over W, then we argue first that case 1 is impossible when considering $\mathbb{P}_2$ as a 1-allowable forcing over $W[G^1]$, where as G^1 is $\mathbb{P}^1$-generic as before.

Indeed, assume for a contradiction that case 1 must be applied at stage β, then, by definition,

$$\begin{aligned} W[G^1][G^2_\beta] \models \exists \mathbb{Q}(&\mathbb{Q} \text{ is 0-allowable with} \\ &\text{respect to } E \text{ and some } F' \\ &\wedge \mathbb{Q} \Vdash x \in A_m(y)). \end{aligned}$$

But then $\mathbb{P}^1 \times \mathbb{Q}$ is such that there is a condition $p^1 \in \mathbb{P}^1 \cap G^1$ which forces that $p^1 \Vdash C^{\mathbb{P}^1} \cap C^{\mathbb{Q}} = \varnothing$. So cutting $\mathbb{P}^1 \times \mathbb{Q}$ below $(p^1, 1)$ will yield a 0-allowable forcing $(\mathbb{P}^1 \times \mathbb{Q})_{\leqslant (p^1,1)}$ such that

$$\begin{aligned} W[G^2_\beta] \models ((&\mathbb{P}^1 \times \mathbb{Q})_{\leqslant (p^1,1)} \text{ is 0-allowable with} \\ &\text{respect to } E \text{ and some } F' \\ &\wedge (\mathbb{P}^1 \times \mathbb{Q})_{\leqslant (p^1,1)} \Vdash x \in A_m(y)). \end{aligned}$$

So we are in case 1 as well when working over $W[G^2_\beta]$, which is a contradiction to our assumption.

Now we need to argue that we are in case 2, when working over $W[G^1][G^2_\beta]$ in the definition of $\mathbb{P}^2$. We can argue in a very similar way than before. So suppose we are at stage β of our iteration and we work over the model $W[G^1][G^2_\beta]$ and case 1 does not apply, which we know already. The universe $W[G^2_\beta]$ thinks that there is $\mathbb{Q}$ which is 0-allowable and which forces $x \in A_k(y)$. Just as before we would like to form $\mathbb{P}^1 \times \mathbb{Q}$ in the model $W[G^2_\beta]$ which will force $x \in A_k(y)$, therefore witnessing that we are in case 2, which is what we want to finish the proof. Again, as before, there is no guarantee that $\mathbb{P}^1$ and $\mathbb{Q}$ have non-intersecting coding areas, so we need to apply again the shifting argument from above. As there we can argue that there is in fact an allowable $\mathbb{Q}$ such that $C^{\mathbb{Q}} \cap C^{\mathbb{P}^1} = \varnothing$ and such that $\mathbb{Q} \Vdash x \in A_k(y)$. Now this $\mathbb{Q}$ can be taken to form a product with $\mathbb{P}^1$, which witnesses that we are in case 2 when defining $\mathbb{P}^2$ at stage β over the model $W[G^2_\beta]$ so this proves the claim for case 2 in the definition of allowable forcings.

Finally, if at stage β, case 3 applies when considering $\mathbb{P}_2$ as a 1-allowable forcing with respect to E over $W[G^2_\beta]$, we claim that we must be in case 3 as well, when defining $\mathbb{P}_2(\beta)$ over $W[G^1][G^2_\beta]$.

If not, then we would be in case 1 or 2 when defining $\mathbb{P}^2(\ \beta)$ over $W[G^1][G^2_\beta]$. Assume without loss of generality that we were in case 1, then there would be a 0-allowable $\mathbb{Q} \in W[G^1][G^2_\beta]$ such that $\mathbb{Q} \Vdash x \in A_m(y)$. Again, via shifting the coding areas of $\mathbb{Q}$ we can assume that the forcing $\mathbb{P}^1 * \mathbb{Q}$ in $W[G^2_\beta]$ is allowable and witnesses that we are in fact in case 1

as well when defining $\mathbb{P}^2(\beta)$ over $W[G^2_\beta]$ which is a contradiction to our assumption that we are in case 3. This shows that we must be in case 3 as well, and the claim, and hence the lemma is finally proved for $\alpha = 1$.

We shall argue now that the Claim is true for $\alpha + 1$-allowable forcings provided we know that it is true for α-allowable forcings. Again we can focus on the case when $\alpha + 1$ allowable forcings are obtained via enlarging E_0, as enlarging E_1 just means to avoid certain coding forcings, which is trivial to be closed under products. We shall show the claim via induction on β. The proof itself is almost identical to the case $\alpha = 1$. As there, we can shift around coding areas using isomorphisms of $(\prod_{\omega_1} \mathbb{C}(\omega_1))^L$ to always find the relevant α-allowable forcings whose coding areas are disjoint. All the work we have to do is replace 0-allowable with α-allowable.

This ends the proof of the claim and so we have shown the lemma. $\square$

6 Ideas for the proof

This section will be used to briefly explain the set up and the structure of the proof of the main theorem. There are two goals we aim to accomplish. First, we want to force $\mathbf{\Sigma^1_3}$-separation. For this, we will use an ω_1-length iteration. We list all Σ^1_3-formulas in two free variables $(\varphi_n(v_0, v_1) \mid n \in \omega)$. We shall use a bookkeeping device which enumerates simultaneously in an ω_1-length list all pairs of natural numbers $(m, k) \in \omega^2$ and (names of) reals $\dot{y}$. These objects $m, k, \dot{y}$ correspond to pairs of $\mathbf{\Sigma^1_3}$-sets $A_m(\dot{y})$ and $A_k(\dot{y})$ (the (name of a) real $\dot{y}$ serves as a parameter in the k-th and m-th Σ^1_3-formula φ_m and φ_k) we want to separate.

At the same time we want to create a universe over which there are two (lightface) Π^1_3-sets B_0 and B_1 which we will design in such a way, that no (boldface) pair of $\mathbf{\Pi^1_3}$-sets exists, which reduces B_0 and B_1.

We settle to work towards $\mathbf{\Sigma^1_3}$-separation on the odd stages of our iteration, whereas we work towards a failure of Π^1_3-reduction on the even stages of the iteration. The iteration itself will consist of the coding forcings $\mathrm{Code}(x, y, m, k, i, \eta)$ applied over L to make certain reals of the form (x, y, m, k) to satisfy our two Σ^1_3-formulas $\Phi_0(x, y, m, k)$ or $\Phi_1(x, y, m, k)$. The final goal is that for any fixed pair of natural numbers m, k, and any parameter $y \in \omega^\omega$, there is a real parameter $R_{y,m,k}$ and a fixed Σ^1_3 formula σ such that the sets

$$D^0_{y,m,k}(R_{y,m,k}) := \{x \mid \Phi_0(x, y, m, k) \wedge \sigma(x, R_{y,m,k})\}$$

and

$$D^1_{y,m,k}(R_{y,m,k}) := \{x \mid \Phi_1(x, y, m, k) \wedge \sigma(x, R_{y,m,k})\}$$

will become the separating sets for the pair of $\Sigma^1_3(y)$-definable sets $A_m(y)$ and $A_k(y)$, i.e. $A_m(y) \subset D^0_{y,m,k}, A_k(y) \subset D^1_{y,m,k}$ and $D^0_{y,m,k}(R_{y,m,k}) \cup D^1_{y,m,k}(R_{y,m,k}) = \omega^\omega$ and $D^0_{y,m,k}(R_{y,m,k}) \cap D^1_{y,m,k}(R_{y,m,k}) = \varnothing$.

At the same time we need to work towards a failure of Π^1_3-reduction which will be done on the even stages of the iteration. We aim to accomplish the failure of Π^1_3-reduction via exhibiting two Π^1_3-sets (lightface) B_0 and B_1 which are chosen in such a way that the question of whether some real x is in B_0 or B_1 can be changed using the coding forcings Code without interfering with the coding forcings we have to use in order to work for the $\mathbf{\Sigma^1_3}$-separation. This freedom will be used to define our ω_1-length iteration of coding forcings so that eventually B_0 and B_1 can not be reduced by any pair of (boldface) $\mathbf{\Pi^1_3}$-sets, thus yielding a slightly stronger failure than just a failure of $\mathbf{\Pi^1_3}$-reduction.

7 The first step of the iteration

We let

$$\vec{\varphi} := (\varphi_n(v_0, v_1) \mid n \in \omega)$$

be a fixed recursive list of the Σ^1_3-formulas in two free variables. We allow that v_0 or v_1 actually do not appear in some of the φ_n's, so our list also contains all Σ^1_3-formulas in one free variable. We use ψ_n to denote $\neg\varphi_n$, i.e. ψ_n is the n-th Π^1_3-formula in the recursive list of Π^1_3-formulas induced by $\vec{\varphi}$.

We fix two Σ^1_3-formulas φ_a, φ_b which provably have non-empty intersection, e.g. $\varphi_a(v_0) = \exists v_0(v_0 = v_0)$ and $\varphi_b = \exists v_0(v_0 = v_0 \wedge v_0 = 1)$. As a consequence, we need not to work towards separating A_a and A_b, thus coding forcings of the form $\mathrm{Code}(\dot{x}, a, b, i, \eta)$ for any (name of a) real $\dot{x}$, for a, b our fixed natural numbers and for $i \in \{0, 1\}$ can be used freely in our definition of the iteration to come.

Next we assume for notational simplicity that $\neg\varphi_0$ and $\neg\varphi_1$, i.e. the negation of the first and the negation of the second formula in our list look like this:

$$\neg\varphi_0(v_0) = \psi_0(v_0) = \neg\Phi_0(v_0) = \text{“}(v_0, a, b) \text{ is not coded into the } \vec{S}^0\text{-sequence”}$$
$$\neg\varphi_1(v_0) = \psi_1(v_0) = \neg\Phi_1(v_0) = \text{“}(v_0, a, b) \text{ is not coded into the } \vec{S}^1\text{-sequence”}$$

Recall [6] that both Φ_0 and Φ_1 are Σ^1_3, hence $\psi)_i$ is Π^1_3. The resulting Π^1_3=sets will be defined like this:

$$B_0 = \{x \in \omega^\omega \mid \psi_0(x)\}$$
$$B_1 = \{x \in \omega^\omega \mid \psi_1(x)\}$$

Note that for any given real $x \in \omega^\omega$, we can manipulate the truth value of $\psi_0(x)$ and $\psi_1(x)$ via using the coding forcings $\mathrm{Code}(x, a, b, 0, \eta)$ and $\mathrm{Code}(x, a, b, 1, \eta)$ respectively from true to false (and once false it will

[6] For a definition of Φ_i look right above Lemma 4.5

remain false because of upwards absoluteness of Σ^1_3-formulas). This in particular will not interfere with the yet to be defined procedure of forming the α_β-allowable forcings, we will need in order to force $\mathbf{\Sigma^1_3}$-separation. Thus we gain some amount of flexibility in how we can make the sets B_0 and B_1 behave. We will use this to diagonalise against all possible $\mathbf{\Pi^1_3}$-sets B_m, B_k in such a way that none of those can reduce B_0 and B_1. This ensures the failure of Π^1_3-reduction.

8 Towards $\mathbf{\Sigma^1_3}$-separation

We are finally in the position to define the iteration which will yield a universe of $\mathbf{\Sigma^1_3}$-separation and a failure of Π^1_3-reduction.

The iteration we are about to define inductively will be an allowable iteration, whose tails are α-allowable and α-increases along the iteration. We start with fixing a bookkeeping function

$$F : \omega_1 \to H(\omega_1)$$

which visits every element cofinally often. The role of F is to list all the quadruples of the form $(\dot{x}, \dot{y}, m, k)$, where $\dot{x}, \dot{y}$ are names of reals in the forcing we already defined, and m and k are natural numbers which represent Σ^1_3-formulas or Π^1_3-formulas with two free variables, cofinally often. Assume that we are at stage $\beta < \omega_1$ of our iteration. By induction we will have constructed already the following list of objects.

- An ordinal $\alpha_\beta \leqslant \beta$ and a set $E_{\alpha_\beta} = E^0_{\alpha_\beta} \cup E^1_{\alpha_\beta}$ which is of the form $\{(\dot{y}_\eta, \dot{m}_\eta, \dot{k}_\eta) : \eta < \alpha_\beta\} \cup \{(\dot{x}_\eta, i_\eta) : \eta < \alpha_\beta\}$, where $\dot{y}_\eta, \dot{x}_\eta$ are $\mathbb{P}_\beta$-names of a reals, $\dot{m}_\eta, \dot{k}_\eta$ are names of natural numbers and i_η a name of a set in $\{0, 1\}$. As a consequence, for every bookkeeping function F' and every generic filter G_β we do have a notion of η-allowable relative to E and F' over $W[G_\beta]$.

- We assume by induction that for every $\eta < \alpha_\beta$, if $\beta_\eta < \beta$ is the η-th stage in $\mathbb{P}_\beta$, where we add a new member to E_{α_β}, then $W[G_{\beta_\eta}]$ thinks that the $\mathbb{P}_{\beta_\eta \beta}$ is η-allowable with respect to $E_{\alpha_\beta} \restriction \eta$.

- If $(\dot{y}_\eta, \dot{m}_\eta, k_\eta) \in E_{\alpha_\beta}$, then we set again β_η to be the η-th stage in $\mathbb{P}_\beta$ such that this new member to E_{α_β} is added. In the model $W[G_{\beta_\eta}]$, we can form the set of reals R_η which were added so far by the use of a coding forcing in the iteration up to stage β_η, and which witness $\Psi_i(x, y, m, k)$ holds for some (x, y, m, k). So in $W[G_\beta]$,

$$R_\eta := \{R \mid R \text{ is the generic filter for a factor of } \mathbb{P}_\beta \text{ of the form } \mathrm{Code}(x, y, m, k)\}$$

Note that R_η is a countable set of reals and can therefore be identified with a real itself, which we will do. The real R_η is an error term and indicates the set of coding areas we must avoid when expecting correct codes, at least for the codes which contain $\dot{y}_\eta, \dot{m}_\eta$ and k_η.

Assume that β is odd, $F(\beta) = (\dot{x}, \dot{y}, \dot{m}, \dot{k})$, assume that $\dot{x}$, $\dot{y}$ are $\mathbb{P}_\beta$-names for reals. Assume that G_β is a $\mathbb{P}_\beta$-generic filter over W. Let x, y, m, k denote the evaluations of the names using G_β. We let $m, k \in \omega$ correspond to the Σ^1_3-formulas $\varphi_m(v_0, v_1)$ and $\varphi_k(v_0, v_1)$. We turn to the forcing $\mathbb{P}(\beta)^{G_\beta}$ we want to define at stage β in our iteration. Again we distinguish several cases.

(A) Assume that $W[G_\beta]$ thinks that there is an α_β-allowable forcing $\mathbb{Q}$ relative to E_{α_β} and some F' such that

$$\mathbb{Q} \Vdash \exists z(z \in A_m(y) \cap A_k(y)).$$

Then we pick the $<$-least such forcing, where $<$ is some previously fixed wellorder. We denote this forcing with $\mathbb{Q}_1$ and use

$$\mathbb{P}(\beta)^{G_\beta} := \mathbb{Q}_1.$$

We do not change R_β at such a stage.

(B) Assume that (A) is not true.

(i) Assume however that there is an α_β-allowable forcing $\mathbb{Q}$ in $W[G_\beta]$ with respect to E_{α_β} such that

$$\mathbb{Q} \Vdash x \in A_m(y).$$

Then we set

$$\dot{\mathbb{Q}}_\beta^{G_\beta} = \mathbb{P}(\beta)^{G_\beta} := \mathrm{Code}(x, y, m, k, 0, \eta),$$

where η is the least ordinal for which the freshness condition holds at stage β. In that situation, we enlarge the E-set as follows. We let $(y, m, k) =: (y_{\alpha_\beta}, m_{\alpha_\beta}, k_{\alpha_\beta})$ and

$$E^{G_{\beta+1}}_{\alpha_\beta+1} := E^{G_\beta}_{\alpha_\beta} \cup \{(y, m, k)\}.$$

Further, if we let r_ζ be the real which uniquely defines the generic filter for $\mathrm{Code}(x_\zeta, y_\zeta, m_\zeta, k_\zeta, i_\zeta, \eta_\zeta)$ at stage ζ of the iteration which witnesses $\Psi_i(x_\zeta, y_\zeta, m_\zeta, k_\zeta)$ of some quadruple $(x_\zeta, y_\zeta, m_\zeta, k_\zeta)$. Then we collect all the countably many such reals r_ζ we have generically added so far in our iteration up to stage β and code them into one real R and let

$$R_{\alpha_\beta+1} := R.$$

(ii) Assume that (i) is wrong, but there is an α_β-allowable forcing $\mathbb{Q}$ with respect to E_{α_β} in $W[G_\beta]$ such that

$$\mathbb{Q} \Vdash x \in A_k(y).$$

Then we set

$$\mathbb{P}(\beta)^{G_\beta} := \mathrm{Code}(x, y, m, k, 1, \eta).$$

Where η is the least ordinal for which the freshness condition at β holds. In that situation, we enlarge the E-set as follows. We let the new $E^{G_{\beta+1}}$ value $(y_{\alpha_\beta}, m_{\alpha_\beta}, k_{\alpha_\beta})$ be (y, m, k) and

$$E^{G_{\beta+1}}_{\alpha_\beta+1} := E^{G_\beta}_{\alpha_\beta} \cup \{(y, m, k)\}.$$

Further, if we let r_ζ be the real which is added by the coding forcing $\mathrm{Code}(x_\zeta, y_\zeta, m_\zeta, k_\zeta, i_\zeta, \eta_\zeta)$ at stage ζ of the iteration which witnesses $\Psi_i(x_\zeta, y_\zeta, m, k)$. Then we collect all the countably many such reals r_ζ we have added so far in our iteration up to stage β and put them into one set R and let

$$R^{G_{\beta+1}}_{\alpha_\beta+1} := R.$$

(iii) If neither (i) nor (ii) is true, then there is no α_β-allowable forcing $\mathbb{Q}$ with respect to E_{α_β} which forces $x \in A_m(y)$ or $x \in A_k(y)$, and we set

$$\mathbb{P}(\beta)^{G_\beta} := \mathrm{Code}(x, y, m, k, 1, \eta).$$

Further, if we let r_ζ be the real which is added by the coding forcing $\mathrm{Code}(x_\zeta, y_\zeta, m_\zeta, k_\zeta, i_\zeta, \eta_\zeta)$ at stage $\zeta \leqslant \beta$ of the iteration which witnesses Ψ_i of some quadruple $(x_\zeta, y_\zeta, m_\zeta, k_\zeta)$. Then we collect all the countably many such reals we have added so far in our iteration up to stage β and put them into one set R and let

$$R^{G_{\beta+1}}_{\alpha_\beta+1} := R.$$

Otherwise we force with the trivial forcing.

9 Towards a failure of Π^1_3-reduction

Assume that $\beta < \omega_1$ is an even stage of our iteration. Our induction hypothesis includes that we have created already the iteration $\mathbb{P}_\beta^{G_\beta}$ up to stage β, and that we defined the notion of α_β-allowable forcings for an ordinal $\alpha_\beta < \omega_1$. Assume that $F(\beta) = (\dot{m}, \dot{k}, \dot{y})$, where $\dot{m}, \dot{k}$ are $\mathbb{P}_\beta$-name for integers and $\dot{y}$ is a $\mathbb{P}_\beta$-name of a real number. We consider the following cases:

9.1 Case 1

We first assume that, working over $W[G_\beta]$, there is a further α_β-allowable forcing $\mathbb{P} \in W[G_\beta]$ such that $\mathbb{P}$ adds two reals x_0 and x_1 and such that for a $\mathbb{P}$-generic filter G over $W[G_\beta]$, $W[G_\beta][G]$ satisfies:

1. $x_0 \in B_m \backslash B_k$ and (x_0, y, a, b) is neither coded into $\vec{S}^0$ nor $\vec{S}^1$.
2. $x_1 \in B_k \backslash B_m$ and (x_1, y, a, b) is neither coded into $\vec{S}^0$ nor $\vec{S}^1$.

In that situation we first use the $<_L$-least $\mathbb{P}_\beta$-name of such a forcing $\mathbb{P}$ to add the reals x_0 and x_1. After forcing with $\mathbb{P}$ we code x_0 and x_1 both into $\vec{S}^1$ via forcing with $\mathrm{Code}(x_0, a, b, 1, \eta_0) \times \mathrm{Code}(x_1, a, b, 1, \eta_1)$ for two distinct ordinals η_0 and η_1 both satisfying the freshness condition.

The two-step forcing

$$\mathbb{P} * (\mathrm{Code}(x_0, a, b, 1, \eta_0) \times \mathrm{Code}(x_1, a, b, 1, \eta_1)) =: \mathbb{P}^{G_\beta}(\beta)$$

is the forcing we use at stage β in the situation of case 1. We also change the notion of α_β-allowable to $\alpha_\beta + 1$-allowable which is defined to be α_β-allowable together with the additional demand to neither use the forcing $\mathrm{Code}(x_0, a, b, 0, \eta)$ nor $\mathrm{Code}(x_1, a, b, 0, \eta)$ for any $\eta < \omega_1$. In other words we let $E_{\alpha_\beta+1} := E_{\alpha_\beta} \cup \{(x_0, 0), (x_1, 0)\}$. Note that this choice ensures that x_0 and x_1 will both be elements of B_0 in all outer $\alpha_\beta + 1$-allowable extensions.

The choice of $\mathbb{P}(\beta) = \dot{\mathbb{Q}}_\beta$ and $\alpha_\beta+1$-allowability will ensure that $B_m(y), B_k(y)$ can not reduce B_0 and B_1 in all possible $\alpha_\beta+1$-generic extensions of $L[G_{\beta+1}]$ as we shall see now.

Lemma 9.1. *The sets $B_m(y)$ and $B_k(y)$ can not reduce B_0 and B_1 in all outer models of $W[G_{\beta+1}]$ which are obtained by a further $\alpha_\beta + 1$-allowable forcing.*

Proof. We shall consider three subcases to prove the lemma. We work in $M \supset L[G_{\beta+1}]$ where M is an outer model obtained by a further $\alpha_\beta + 1$-allowable forcing.

Case 1a: First we assume that $x_0 \in B_m(y) \backslash B_k(y)$ and $x_1 \in B_k(y) \backslash B_m(y)$ still holds in M. In this situation neither $B_m(y) \subset B_1$ nor $B_k(y) \subset B_1$ can hold as witnessed by x_0 and x_1. In particular, $B_m(y)$ and $B_k(y)$ can not reduce B_0 and B_1. Note that this will remain true in all further outer models obtained by an additional $\alpha_\beta + 1$-allowable forcing as long as $x_0 \in B_m(y) \backslash B_k(y)$ and $x_1 \in B_k(y) \backslash B_m(y)$. If x_i will drop out of $B_m(y)$ or $B_k(y)$ in some $\alpha_\beta + 1$-allowable extension, then case 1b and case 1c will apply.

Case 1b: We assume that $x_0 \in B_m(y) \backslash B_k(y)$ but $x_1 \notin B_m(y) \cup B_k(y)$ holds in M. In this situation we can not have $B_0 \cup B_1 = B_m(y) \cup B_k(y)$ as $x_1 \in B_0$

and $x_1 \notin B_m(y) \cup B_k(y)$. Note that $x_1 \in B_0$ will remain true in outer $\alpha_\beta+1$-allowable models by our choice of $E_{\alpha_\beta+1}$ and $x_1 \notin B_m(y) \cup B_k(y)$ by the upwards absoluteness of the Σ^1_3-formulas $\neg\psi_m$ and $\neg\psi_k$. As a consequence, $B_m(y)$ and $B_k(y)$ can not reduce B_0 and B_1 in M.

Case 1c: In the dual case we assume that $x_1 \in B_m(y) \backslash B_k(y)$ but $x_0 \notin B_m(y) \cup B_k(y)$ holds in M to derive, as above, that $B_0 \cup B_1 \neq B_m(y) \cup B_k(y)$.

Case 1d: If x_0 and x_1 are both not in $B_m(y)$ and $B_k(y)$, then again $B_m(y) \cup B_k(y) \neq B_0 \cup B_1$.

□

9.2 Case 2

In the second case, we assume that case 1 does not apply. As a consequence, whenever we work over $W[G_\beta]$ and apply a further α_β-allowable forcing $\mathbb{P}$ which adds two reals $x_0 \neq x_1$ and does neither have $\mathrm{Code}(x_0, a, b, i, \eta)$ nor $\mathrm{Code}(x_1, a, b, i, \eta)$ for some $\eta < \omega_1$ and for $i \in \{0, 1\}$ as a factor in the iteration, then x_0, x_1 will not satisfy that $x_0 \in B_m(y) \backslash B_k(y)$ and $x_1 \in B_k(y) \backslash B_m(y)$.

Again, we shall split into subcases:

Case 2a: There is an α_β-allowable $\mathbb{P}$ and a $G \subset \mathbb{P}$ such that in $W[G_\beta][G]$ there are $x_0 \neq x_1$ such that $x_0, x_1 \in B_m \backslash B_k$. In this situation, we force over $W[G_\beta][G]$ with $\mathrm{Code}(x_0, a, b, 1, \eta_0) \times \mathrm{Code}(x_1, a, b, 0, \eta_1)$ for two ordinals η_0, η_1 for which the freshness condition holds. Let $H_0 \times H_1 \subset \mathrm{Code}(x_0, a, b, 1) \times \mathrm{Code}(x_1, a, b, 0)$ be a $W[G_\beta][G]$-generic filter; and define $G_{\beta+1} := G_\beta * G * (H_0 \times H_1)$. We also define $\alpha_\beta + 1$-allowable via enlarging $E_{\alpha_\beta+1} := E_{\alpha_\beta} \cup \{(x_0, 0), (x_1, 1)\}$. That is we introduce the additional rule that $\mathrm{Code}(x_0, a, b, 0, \zeta_0)$ and $\mathrm{Code}(x_1, a, b, 1, \zeta_1)$, for any $\zeta_0, \zeta_1 < \omega_1$ will not be used anymore as factors of the iterations. Note that as a consequence of this, $x_0 \in B_0$ and $x_1 \in B_1$ will remain true in all $\alpha_\beta + 1$-allowable generic extensions of $W[G_{\beta+1}]$. Then we argue as follows:

Lemma 9.2. *Let M be an outer model of $W[G_{\beta+1}]$ obtained via an $\alpha_\beta + 1$-allowable forcing. Then $B_m(y)$ and $B_k(y)$ can not reduce B_0 and B_1 over M.*

Proof. We split into several cases. Assume first that in $M \supset W[G_{\beta+1}]$, $x_0, x_1 \in B_m(y)$ still holds true. Then, as $x_0 \in B_0 \backslash B_1$ and $x_1 \in B_1 \backslash B_0$, the set $B_m(y)$ can neither reduce B_0 nor B_1. Indeed x_0 witnesses that $B_m(y)$ is not a subset of B_1 and x_1 witnesses that $B_m(y)$ is not a subset of B_0. If, in a further $\alpha_\beta + 1$-allowable extension of $M \supset W[G_{\beta+1}]$,

$x_0 \notin B_m(y)$ or $x_1 \notin B_m(y)$ then this will be dealt with in the next subcases.

If in $M \supset W[G_{\beta+1}]$, $x_0 \notin B_m(y)$, yet $x_1 \in B_m(y)$ then $x_0 \notin B_m(y) \cup B_k(y)$ and $x_0 \in B_0$ in all $\alpha_\beta + 1$-allowable generic extensions of M. So $B_m(y) \cup B_k(y) \neq B_0 \cup B_1$ and $B_m(y), B_k(y)$ can not reduce B_0, B_1. The same argument works if $x_1 \notin B_m(y)$ and $x_0 \in B_m(y)$.

If in M, $x_0, x_1 \notin B_m(y)$ anymore, then again $B_m(y), B_k(y)$ can not reduce B_0, B_1.

□

Case 2b: The dual situation in which we can force $x_0 \neq x_1$ such that $x_0, x_1 \in B_k(y) \backslash B_m(y)$ is dealt with in the analogous way.

Case 2c: If neither 2a nor 2b are true, then one can not force two distinct reals x_0, x_1 into $B_m(y) \backslash B_k(y)$ with α_β-allowable forcings which do not contain $\mathrm{Code}(x_0, a, b, i.\eta)$ or $\mathrm{Code}(x_1, a, b, i, \eta)$ for $i \in \{0, 1\}$ and $\eta < \omega_1$. And the same holds true for $B_k(y) \backslash B_m(y)$. But it is straightforward to use an α_β-allowable forcing $\mathbb{P}$ over $W[G_\beta]$ which will add $\aleph_1$-many new reals $(z_i \mid i < \omega_1)$ and neither $\mathrm{Code}(z_i, a, b, 0, \eta)$ nor $\mathrm{Code}(z_i, a, b, 1, \eta)$ is a factor of $\mathbb{P}$. We force with such a $\mathbb{P}$, let $G \subset \mathbb{P}$ be generic over $W[G_\beta]$ and let $W[G_{\beta+1}] = W[G_\beta][G]$. We set $\alpha_\beta + 1$-allowable forcing to be α_β-allowable with the additional constraint that neither $\mathrm{Code}(z_i, a, b, 0, \eta)$ nor $\mathrm{Code}(z_i, a, b, 1, \eta)$ must be used for any $\eta < \omega_1$.

Lemma 9.3. *If M is an outer model of $W[G_{\beta+1}]$, obtained with a further $\alpha_\beta + 1$-allowable forcing. Then $B_m(y)$ and $B_k(y)$ can not reduce B_0, B_1.*

Proof. For every $\alpha_\beta + 1$-allowable extension M of $W[G_{\beta+1}]$, $\forall i < \omega_1 (z_i \in B_0 \cup B_1)$ must hold in M, yet for any pair $z_i \neq z_j$, $M \models z_i, z_j \in B_m(y) \cap B_k(y)$ or $M \models z_i, z_j \notin (B_m(y) \cup B_k(y))$. In both cases, z_i, z_j witness that $B_m(y)$ and $B_k(y)$ can not reduce B_0, B_1 over M.

□

To summarize: in both cases we defined an extension $W[G_{\beta+1}]$ of $W[G_\beta]$, and the notion of $\alpha_\beta + 1$-allowable forcings. Additionally we found reals which witness that $B_m(y)$ and $B_k(y)$ can not reduce B_0 and B_1 in all further outer models M of $W[G_{\beta+1}]$ which are obtained with a further $\alpha_\beta + 1$-allowable forcing.

At limit stages β, we use the finite support to define the limit partial order $\mathbb{P}_\beta$ and set $E_{\alpha_\beta} = \bigcup_{\eta<\beta} E_{\alpha_\eta}$. This ends the definition of $\mathbb{P}_{\omega_1}$.

10 Discussion of the resulting universe

We let G_{ω_1} be a $\mathbb{P}_{\omega_1}$-generic filter over W. As $W[G_{\omega_1}]$ is a proper extension of W, ω_1 is preserved. Moreover CH remains true.

A second observation is that for every stage β of our iteration and every $\eta > \beta$, the intermediate forcing $\mathbb{P}_{[\beta,\eta)}$, defined as the factor forcing of $\mathbb{P}_\beta$ and $\mathbb{P}_\eta$, is always an α_β-allowable forcing relative to E_{α_β} and some bookkeeping. This is clear as by the definition of the iteration, we force at every stage β with a α_β-allowable forcing relative to E_{α_β} and α_β-allowable becomes a stronger notion as we increase α_β.

A third observation is, that B_0 and B_1 can not be reduced by any pair of boldface Π^1_3-sets $B_m(y), B_k(y)$. This follows immediately from the definition and the discussion of the iteration on even stages. Indeed if $B_m(y)$ and $B_k(y)$ are arbitrary $\mathbf{\Pi^1_3}$-sets, then, if $\dot{y}$ is some nice name of y, the triple $(m, k, \dot{y})$ will be guessed by the bookkeeping function cofinally often on the even stages. But on the first stage β where it is guessed, we ensured that B_0 and B_1 can not be reduced by $B_m(y)$ and $B_k(y)$. And this remains true for all further $\alpha_\beta + 1$-allowable extensions of $L[G_{\beta+1}]$. As $L[G_{\omega_1}]$ is an $\alpha_\beta + 1$-allowable extension of $L[G_{\beta+1}]$, the assertion follows.

We still need to show that in $L[G_{\omega_1}]$ the $\mathbf{\Sigma^1_3}$-separation property is true. For a pair of disjoint $\Sigma^1_3(y)$-sets, $A_m(y)$ and $A_k(y)$, we consider the least stage β such that there is a $\mathbb{P}_\beta$-name $\dot{z}$ such that $\dot{z}^{G_\beta} = z$ and (z, y, m, k) are considered by F at stage β. We consider the real R_{α_β} which codes up all the reals which witness an instance of $(***)$ for some quadruple (x', y', m', k'). This real R_{α_β} contains the information for all coding areas and quadruples (x', y', m', k') we have coded up so far into $\vec{S}$ in our iteration. As mentioned already, the role of R_{α_β} is that of an error term. We might have added false patterns in our iteration so far, but these false patterns will appear in a coded form in R_{α_β}. Our definitions will ensure that for the pair $A_m(y)$ and $A_k(y)$, modulo R_{α_β}, the set of reals x for which the quadruple (x, y, m, k) is coded into $\vec{S}^0$, and the set of reals x, for which the quadruple (x, y, m, k) is coded into $\vec{S}^1$ will separate $A_m(y)$ and $A_k(y)$:

$$\begin{aligned} x \in D^0_{y,m,k}(R_{\alpha_\beta}) \Leftrightarrow \exists r(L[r, R_{\alpha_\beta}] \models (x, y, m, k) &\text{ can be read off from a code} \\ &\text{written on an } \omega_1\text{-many } \omega\text{-blocks of elements of} \\ &\vec{S}^0 \text{ and the coding area of } (x, y, m, k) \\ &\text{is almost disjoint from each coding area in } R_{\alpha_\beta}). \end{aligned}$$

and

$$x \in D^1_{y,m,k}(R_{\alpha_\beta}) \Leftrightarrow \exists r(L[r, R_{\alpha_\beta}] \models (x, y, m, k) \text{ can be read off from a code written on an } \omega_1\text{-many } \omega\text{-blocks of elements of } \vec{S^1} \text{ and the coding area of } (x, y, m, k) \text{ is almost disjoint from each coding area in } R_{\alpha_\beta}).$$

It follows from the definition of the iteration that for any real parameter y and any $m, k \in \omega$, $D^0_{y,m,k}(R_{\alpha_\beta}) \cup D^1_{y,m,k}(R_{\alpha_\beta}) = \omega^\omega$. Indeed, as our bookkeeping function visits each name of a real in our iteration $\aleph_1$-many times, it will list each real $x \in W[G_{\omega_1}]$ unboundedly often below ω_1. Thus (x, y, m, k) will be coded into $\vec{S^0}$ or $\vec{S^1}$ at stages $\beta' > \beta$, which gives that $D^0_{y,m,k}(R_{\alpha_\beta}) \cup D^1_{y,m,k}(R_{\alpha_\beta}) = \omega^\omega$ The next lemma establishes that the sets are indeed separating.

Lemma 10.1. *In $W[G_{\omega_1}]$, let y be a real and let $m, k \in \omega$ be such that $A_m(y) \cap A_k(y) = \varnothing$. Then there is an real R such that the sets $D^0_{y,m,k}(R)$ and $D^1_{y,m,k}(R)$ partition the reals.*

Proof. Let β be the least stage such that there is a real x such that $F(\beta) = (\dot{x}, \dot{y}, m, k)$ with $\dot{x}^G_\beta = x$, $\dot{y}^G_\beta = y$. Let R be R_{α_β} for R_{α_β} being defined as above. Then, as $A_m(y)$ and $A_k(y)$ are disjoint in $W[G_{\omega_1}]$, by the rules of the iteration, case B must apply at β.

Assume now for a contradiction, that $D^0_{y,m,k}(R)$ and $D^1_{y,m,k}(R)$ do have non-empty intersection in $W[G_{\omega_1}]$. Let $z \in D^0_{y,m,k}(R) \cap D^1_{y,m,k}(R)$ and let $\gamma > \beta$ be the first stage of the iteration which sees that z is in the intersection. Then, by the rules of the iteration and without loss of generality, we must have used case B(i) at the first stage $\delta \geqslant \beta$ of the iteration where (z, y, m, k) is listed by the bookkeeping F, and case B(ii) at stage γ. But this would imply, that at stage δ, there is an α_δ-allowable forcing $\mathbb{Q}_\delta$ with respect to E_{α_δ}, which forces $z \in A_m(y)$, yet at stage $\gamma > \delta$, there is an α_γ-allowable forcing which forces $z \in A_k(y)$. As $\gamma > \delta$, the α_γ-allowable forcing $\mathbb{Q}_\gamma$ which witnesses that we are in case B (ii) at stage γ in our iteration, is also α_δ-allowable. But this means that, over $W[G_\delta]$, the intermediate forcing $\mathbb{P}_{\delta,\gamma}$, which is also α_δ-allowable can be extended to the α_δ-allowable forcing which first uses $\mathbb{P}_{\delta,\gamma}$ and then $\mathbb{Q}_\gamma$ yielding an α_δ-allowable forcing which forces $z \in A_k(y)$.

On the other hand, in $W[G_\delta]$ we do have $\mathbb{Q}_\delta$ which forces $z \in A_m(y)$, as we assumed that at stage δ we are in case B (i). Now the product $\mathbb{Q}_\delta \times (\mathbb{P}_{\delta,\gamma} * \mathbb{Q}_\gamma)$ is α_δ-allowable, and by upwards absoluteness of Σ^1_3-formulas we get that

$$\mathbb{Q}_\delta \times (\mathbb{P}_{\delta,\gamma} * \mathbb{Q}_\gamma) \Vdash z \in A_m(y) \cap A_k(y).$$

But this would mean that at stage δ, we are in case A in the definition of our iteration, which is a contradiction.

□

Lemma 10.2. *In $W[G_{\omega_1}]$, for every pair m, k and every parameter $y \in \omega^\omega$ such that $A_m(y) \cap A_k(y) = \varnothing$ there is a real R such that*

$$A_m(y) \subset D^0_{y,m,k}(R) \wedge A_k(y) \subset D^1_{y,m,k}(R)$$

Proof. The proof is by contradiction. Assume that m, k and y are such that for every real R there is z such that $z \in A_m(y) \cap D^1_{y,m,k}(R)$ or $z \in A_k(y) \cap D^1_{y,m,k}(R)$. We consider the smallest ordinal $\beta < \omega_1$ such that $F(\beta)$ lists a quadruple of the form (x, y, m, k) for which $W[G_{\omega_1}] \models x \in A_m(y) \cap D^1_{y,m,k}$ and let $R = R_{\alpha_\beta}$. As $A_m(y)$ and $A_k(y)$ are disjoint we know that at stage β we were in case B. As x is coded into $\vec{S}^1$ after stage β and by the last Lemma, Case B(i) is impossible at β. Hence, without loss of generality we may assume that case B(ii) applies at β. As a consequence, there is a forcing $\mathbb{Q}_2 \in W[G_\beta]$ which is α_β-allowable with respect to E_{α_β} which forces $\mathbb{Q}_2 \Vdash x \in A_k(y)$. Note that in that case we collect all the reals which witness $(***)$ for some quadruple to form the set R_{α_β}.

As $x \in A_m(y) \cap D^1_{y,m,k}(R)$, we let $\beta' > \beta$ be the first stage such that $W[G_{\beta'}] \models x \in A_m(y)$. By Lemma 5.1, $W[G_\beta]$ thinks that $\mathbb{Q}_2 \times \mathbb{P}_{\beta\beta'}$ is α_β-allowable with respect to E_{α_β}, yet $\mathbb{Q}_2 \times \mathbb{P}_{\beta\beta'} \Vdash x \in A_m(y) \cap A_k(y)$. Thus, at stage β, we must have been in case A. This is a contradiction. □

The next lemma will finish the proof of our theorem:

Lemma 10.3. *In $W[G_{\omega_1}]$, if $y \in \omega^\omega$ is an arbitrary parameter, R a real and m, k natural numbers, then the sets $D^0_{y,m,k}(R)$ and $D^1_{y,m,k}(R)$ are $\Sigma^1_3(R)$-definable.*

Proof. The proof is a standard calculation using the obvious modification of $(***)$ employing R as the set which codes the coding areas a real must avoid to be in $D^0_{y,m,k}(R)$ or $D^1_{y,m,k}(R)$. □

11 Lifting to M_n

The ideas presented can be used to construct a universe in which $\mathbf{\Sigma^1_{n+3}}$-separation can be separated from Π^1_{n+3}-reduction. Its proof is a direct application of the ideas from [7] which can be used to translate the argument for the third level of the projective hierarchy to M_n. As there are no new ideas needed, and the translation works in a very similar manner to [7], we will not go into any details.

12 Open questions

We end with several questions which are related to this article.

Question 1. *Does there exist a universe in which* $\mathbf{\Sigma^1_3}$*-separation holds, but* Π^1_n*-reduction fails for any* $n \geqslant 3$*?*

Note for this question that in our construction of the failure of Π^1_3-reduction, we used Shoenfield absoluteness, or rather the upwards absoluteness of Σ^1_3-formulas, several times. Thus a failure of Π^1_n-reduction would need new ideas and arguments.

Question 2. *Can one force a universe of* $\mathbf{\Sigma^1_4}$*-separation in which* Π^1_4*-reduction fails over just* L*?*

By a classical result of Novikov, for any projective pointclass Γ, it is impossible to have Γ and $\check{\Gamma}$-reduction simultaneously. The case for separation is still unknown.

Question 3. *Can one force a universe where* Σ^1_3*- and* Π^1_3*-separation hold simultaneously?*

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